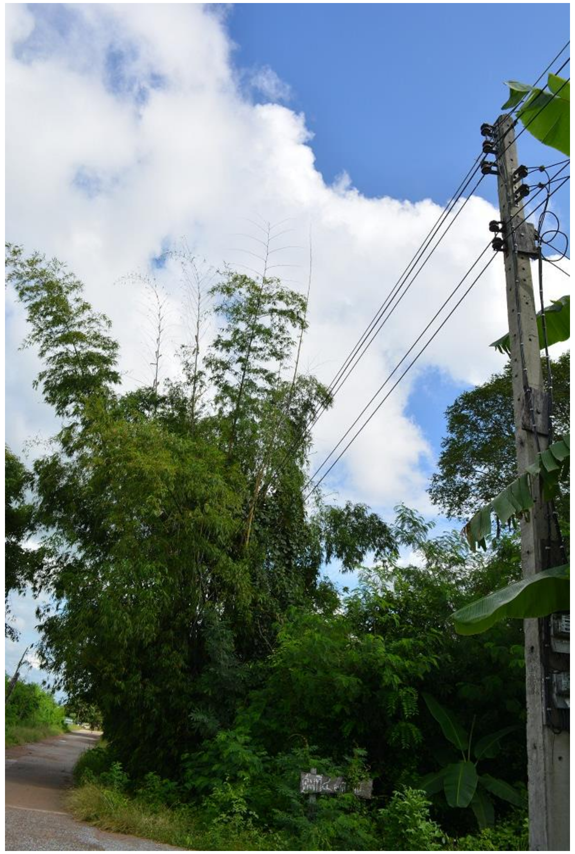
TonmaAI Vegetation Management – Business Solution Manual

September 2024

Contents

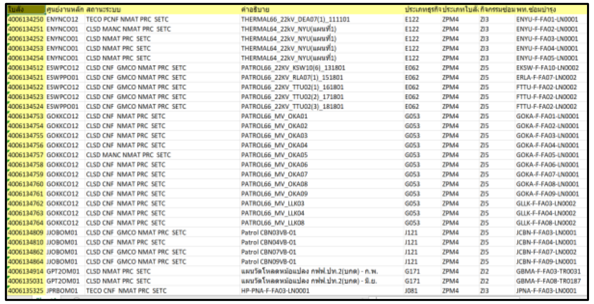
Vegetation Management will help PEA allocate resources based on risk to reduce costs and increase reliability, by moving from reactive decision making based on in-person survey results to state-of-art Advanced Analytics methodologies

Use case	From	To



From

- วางแผนแบบ **Manual Route-based** และขาดการประสานร่วมกันระหว่างเขตในบางกรณี
- ขาดการจัดลำดับความสำคัญของพื้นที่ที่ต้องการการตัดแต่งต้นไม้
- ขาดขั้นตอนมาตรฐานในการตรวจสอบงานของผู้รับเหมา



ใบสั่ง	ศูนย์งานหลัก	สถานะระบบ	คำอธิบาย	ประเภทธุรกิจ	ประเภทใบสั่	กิจกรรมซ่อม	พท.ซ่อมบำรุง
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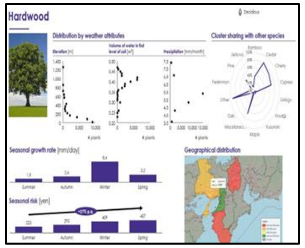
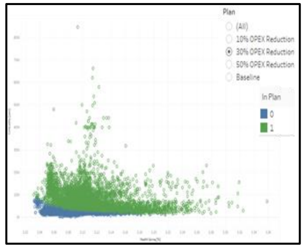
4006135031	GPT2OM01	CLSD NMAT PRC SETC	แผนวัดโหลดหม้อแปลง กฟฟ.ปท.2(บกด) - มิ.ย.	G171	ZPM4	Z12	GBMA-F-FA08-TR0187
4006135325	JPRBOM01	TECO CNF NMAT PRC SETC	HP-PNA-F-FA03-LN0001	J081	ZPM4	Z13	JPNA-F-FA03-LN0001

ค่าใช้จ่ายการดำเนินงานและ ~1,000 ล้านบาท

To

- ปรับปรุงการวางแผนเป็น **Condition/Critically-based** ที่ครอบคลุมทุกเขตโดยใช้ **AI/Advanced Analytics**
- **ปรับระเบียบการตรวจสอบราคากลาง** ให้สอดคล้องกับความหนาแน่นจริง





ค่าใช้จ่ายการดำเนินงานและ 870 ล้านบาท ลดลง 130 ล้านบาท

THB ~106 Mn OPEX savings

Time to stage 4: <6 months	Payback period: TBC	IRR: TBC

PEA PROVINCIAL ELECTRICITY AUTHORITY

Vegetation Management aims to allocate resources based on risk to reduce costs and increase reliability, by moving from reactive decision making based on in-person survey results to state-of-art Advanced Analytics methodologies

	From…	…To
Probability of outage	**No model is leveraged.** Trimming is determined based on the in-person surveys carried out by subdistrict	**Advanced Analytics models** based on Machine Learning, that accurately **predict the probability of outage of each corridor** based on historical events
Consequence of outage		Model considering the importance of each corridor given the **customers connected**, **type of customers** and **load**
Maintenance optimization	**2 surveys are carried out** for the year of trimming in the same place • Survey 1 (*Year Y-1*): **Request budget allocation** for trimming from the District team • Survey 2 (*Few days before PO launch*): **Update the same survey with current density numbers** **Actual trimming** is carried out in the plan's sections the **given year based on last survey**	Corridors to be trimmed yearly are **determined based on an optimization of risk vs cost** allowing to focus resources on areas that are more likely and critical to generate vegetation-related outages to **reduce costs and increase reliability**

เราจะดำเนินการปรับปรุงกระบวนการวางแผนและจัดทำงบตัดต้นไม้แบบครบวงจรตั้งแต่ Data Collection ไปจนถึง Decision-Making Support

PEA data | X Detailed next

1 รวมรวมและคัดเลือกปัจจัยที่มีผลต่อการชำรุดและข้อมูลสำหรับการคำนวณ

- **GIS data**
- **Outage history**
- **Trim history**
- **Trim cost**
- **Weather / climate models**

ปรับปรุงข้อมูลที่จัดเก็บและรูปแบบการเก็บข้อมูล

2 **Corridor Model[1]**

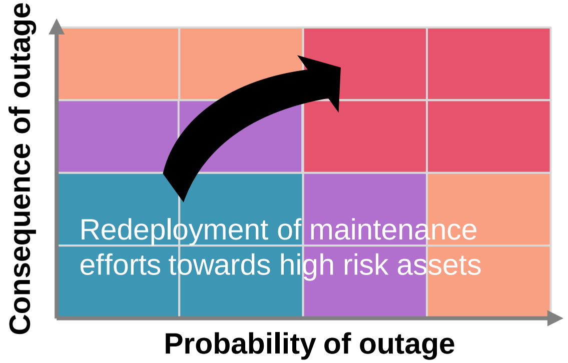
3 การวิเคราะห์ความหนาแน่นของต้นไม้ผ่านภาพดาวเทียม

4 คาดการณ์โอกาสในการทำให้เกิดไฟดับ [2]

งบประมาณในการตัดต้นไม้ [1]

5 คำนวณความสำคัญของแต่ละจุดต่อระบบ[2]

พัฒนาและสร้าง **Risk-based, decision-making model**



Operational constraints Workload, contractor, environmental

6 **Optimization[3]**

แนะนำแผนการเปลี่ยนอุปกรณ์และงบที่ต้อง

- Corridors to be trimmed given max budget or max risk
- Comparison of model output to current plan[4]
- งบประมาณการตันไม้

จัดทำ **Scenario** ตามงบประมาณหรือความเสี่ยง

7 พัฒนา User Interface เพื่อรองรับการอัพโหลดข้อมูล การดึงข้อมูลและการใช้ข้อมูลในการตัดสินใจ

1. Rule based model; 2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming; 4. Comparison consists of: trimming feeders, total cost, density of trees and therefore trimming cost by feeder

Source: McKinsey & Company

Vegetation management has been developed for primary overhead medium voltage lines focusing on PEA Phunphin and PEA Trang Districts

KPIs		Km of lines	Years of outage data	Vegetation-related outages[1]	Data Sources
	Phunphin	2K	4	3.8K	6
	Trang	1K		1.3K	

Objective

Optimize trimming cycle according to the probability of vegetation-related outages and the related consequence of outage depending on the number of downstream customers at risk
Confirm/challenge vegetation density charges by contractor

Location

PEA Phunphin District and PEA Trang, corresponding to Feeder IDs PPA, PPB and TRA

Asset

Primary overhead medium voltage lines. The unit for which outage probability is predictive and therefore the **granularity level is corridor**, defined as the group of overhead lines that roll up to the same nearest upstream protective device

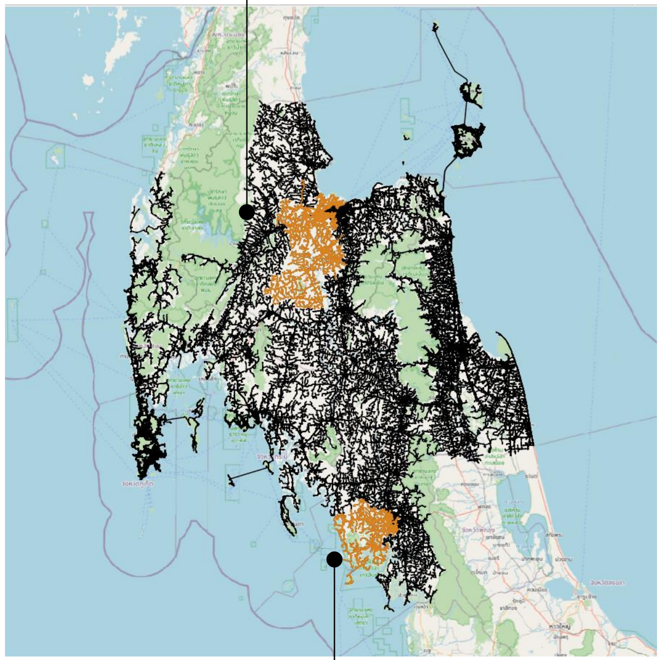
Outage

Unplanned power outage events due to vegetation happened in 2020-2023. For example: tree/branch falling on the line, tree/branch touching the line due to wind or excessive growth

Data

- **Utility GIS data** - structural information of the line and network map
- **Outages** - target variable used to train the outage prediction model
- **Trimming history, climate, remote sensing data** – explicative variables which help predicting the outages

PEA Phunphin



PEA Trang

1 Filtering out 4% of outages due to vegetation-related outages during storms

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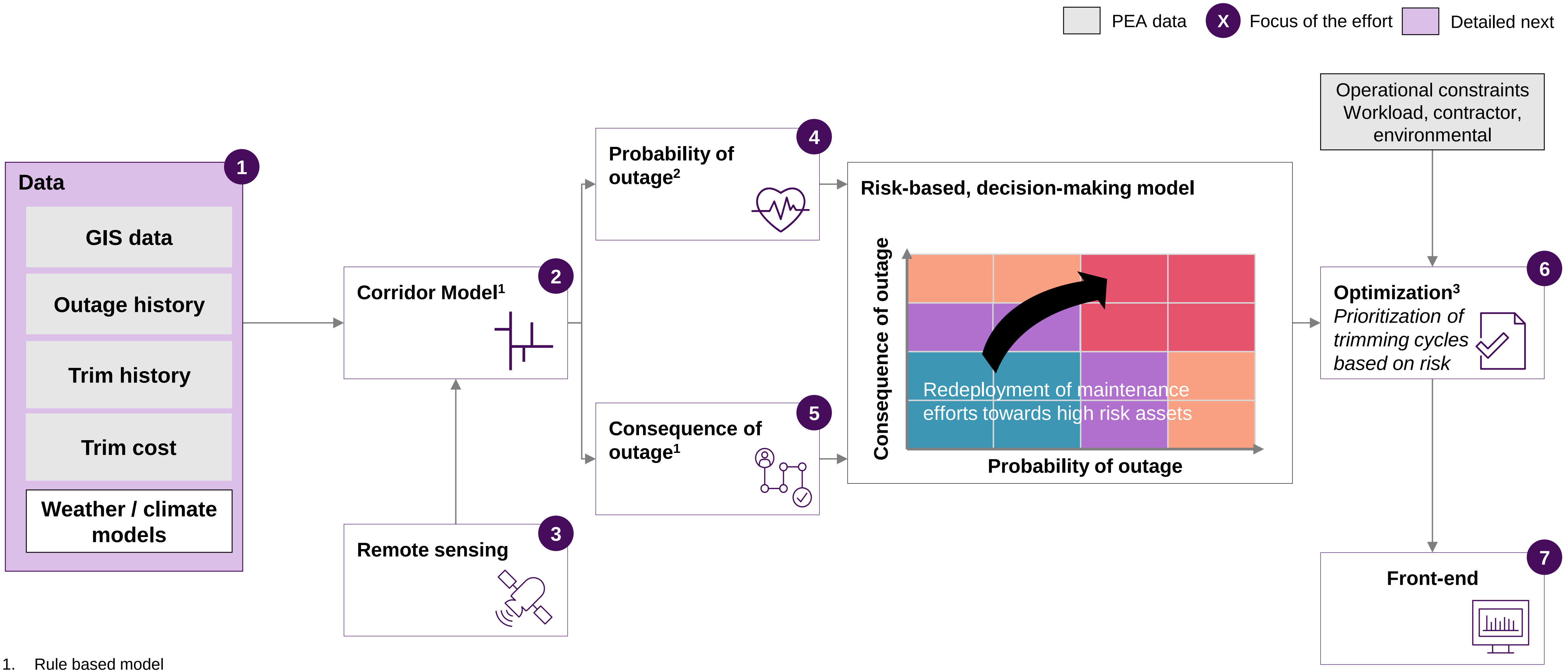

1. Rule based model
2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming

Source: McKinsey & Company

While GIS and outage data quality is high, trimming data requires collecting information from several sources and processing it due to different levels of granularity

Fully available · Partially available · High · Medium · Low · Detailed next

Data Group	Availabity	Description	Status	Comments
1 GIS data	Fully available	All necessary files for creating corridors and extracting poles/line characteristics Phang Nga, Phuket, Krabi, Surat Thani, Nakhon Si Thammarat, Trang	High	**Mapped 99% of primary lines to devices**
2 Outage data	Fully available	Available for 2020 – 2024 Entire PEA lines	High	**Mapped 97% of outages to devices** Vegetation-related outages can be classified as grown into line or fall on-line
2 Trimming history	Partially available	Available for: • 2020-2023: SAP • 2023-2024: MJM Trang and Phunphin Districts	Medium Low granularity in 2020-2023 data in SAP	**SAP system** (2020-2023) – currently manually processing free text description to derive device-level trimming location **MJM system** (2023-2024) – provides point-level location and activity date for trimming performed in 2023 and 2024[2], used as baseline
Weather data[1]	Fully available	Availability for 2020 – 2024 Temperature, winds speed, and precipitation	Medium Good quality but low granularity	Open-source weather data is **only available at weather station level.** Weather stations can be far away from nearest corridor
Remote sensing	Partially available	Satellite images: • Sentinel-2 imagery • META[3] Canopy Height 2018 - 2020	Medium Medium spatial resolution	**Sentinel-2** 10m resolution, 5 days frequency – will be included in the canopy presence and density calculation **Meta**: High correlation (68% R2) between META-calculated density to trees per kilometer on 10 variable-length test line segments
Rate card middle price	Fully available	Trimming coefficient cost by density and types of line – available	High	**Available**

1. Source: Meteostat https://dev.meteostat.net/python/;
2. Planned in 2022-2023
3. Download source: https://registry.opendata.aws/dataforgood-fb-forests/

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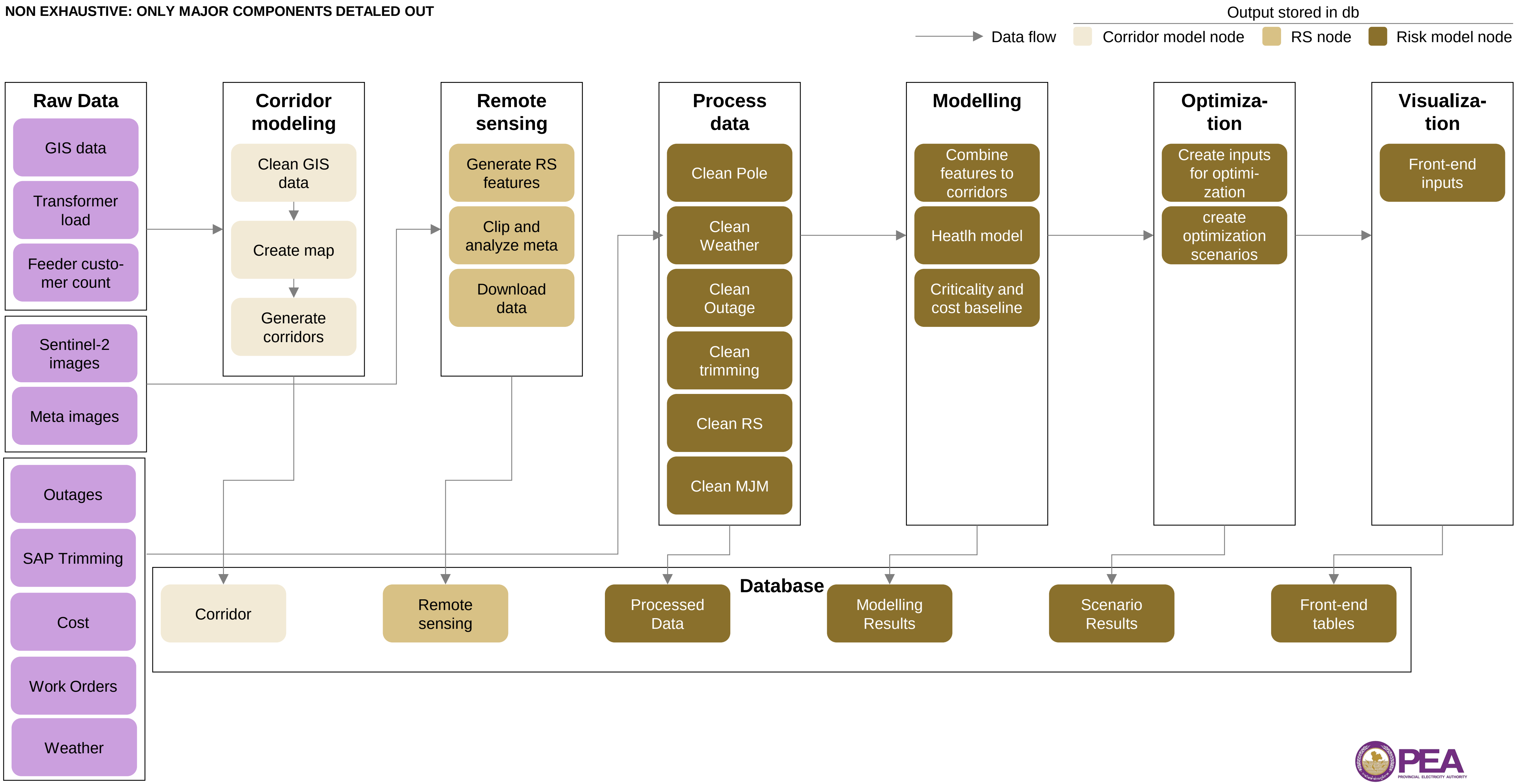
Data Pipelines access database and local folder to generate visuals and scenario results

NON EXHAUSTIVE: ONLY MAJOR COMPONENTS DETALED OUT



1. Assessing GIS poles and overhead lines

Methods for mapping GIS lines

Methods for mapping lines:

1) GIS lines – use the current state GIS lines to build out VegX solution pipeline
2) GIS poles – geospatially map lines to poles to update locations of lines

Key findings:

- GIS lines, and poles can deviate from the ground truth by at 3-15 meters
- MJM lines have the greatest deviation of 3-20 meters

- There is no consistent pattern indicating whether poles, GIS lines, or MJM lines are closer to the ground truth. Sometimes lines are more accurate than poles, and vice versa.

Current model

- Uses GIS lines to identify density bins
- Apply buffer on GIS lines of 10 meters either side for account for offset
- **Future steps…**
  1) Manually correct lines and poles to ground truth imagery
  2) Incorporate unique line to pole identifiers in GIS data to map lines to poles automatically

1. Case 1: GIS lines are closer to ground truth than poles

MJM lines MV lines GIS poles Google satellite ground truth

- Clear pattern in this example of lines and poles being offset
- GIS lines appear closer to ground truth than poles
- **GIS lines** are about 3 meters off from ground truth
- MJM lines are the closest to the ground truth
- GIS poles are about 9 meters off from ground truth

1. Case 2: GIS lines are closer to ground truth than poles

MJM lines   MV lines   GIS poles   Google satellite ground truth

- While the poles show a pattern in their offset, the GIS and MJM lines are still closer to the ground truth
- **GIS lines** are about 1 meters off from ground truth
- MJM lines are about 1 meter off from **GIS lines**
- GIS poles are about 5 meters off from ground truth

1. Case 3: Poles are closer to ground truth than GIS lines

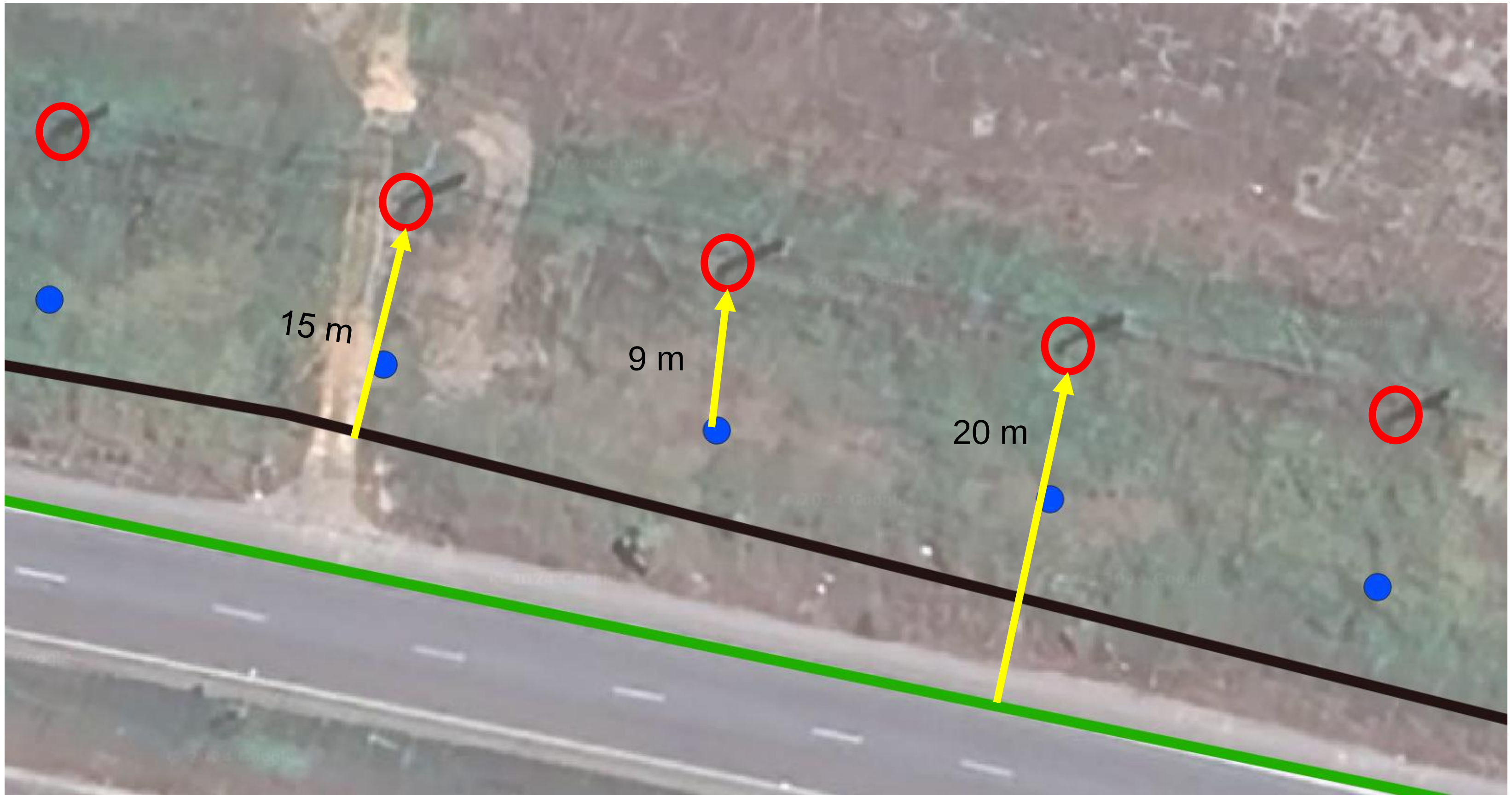
MJM lines   MV lines   GIS poles   Google satellite ground truth

- Poles appear along the road while GIS and MJM lines can be found on the road
- Poles appear closer to ground truth than GIS lines
- GIS line buffer of 10 meters will capture actual vegetation
- **GIS lines** are about 5 meters off from ground truth
- MJM lines are farthest from the ground truth by about 8 meters
- GIS poles are about 2 meters off from ground truth

PEA PROVINCIAL ELECTRICITY AUTHORITY

1. Case 4: Poles are closer to ground truth than GIS lines

MJM lines MV lines GIS poles Google satellite ground truth



- Poles appear closer to ground truth than GIS lines
- While buffer may not capture the ground truth vegetation, in this example there is no clear canopy present
- **GIS lines** are about 15 meters off from ground truth
- MJM lines are farthest from the ground truth by about 20 meters
- GIS poles are about 6 meters off from ground truth

2. SAP data analysis indicate that outages happen in areas trimmed recently, demonstrating trimming tends to be concentrated in risky places and may not be efficient enough in some cases

Corridors with recorded trimmings fail more …

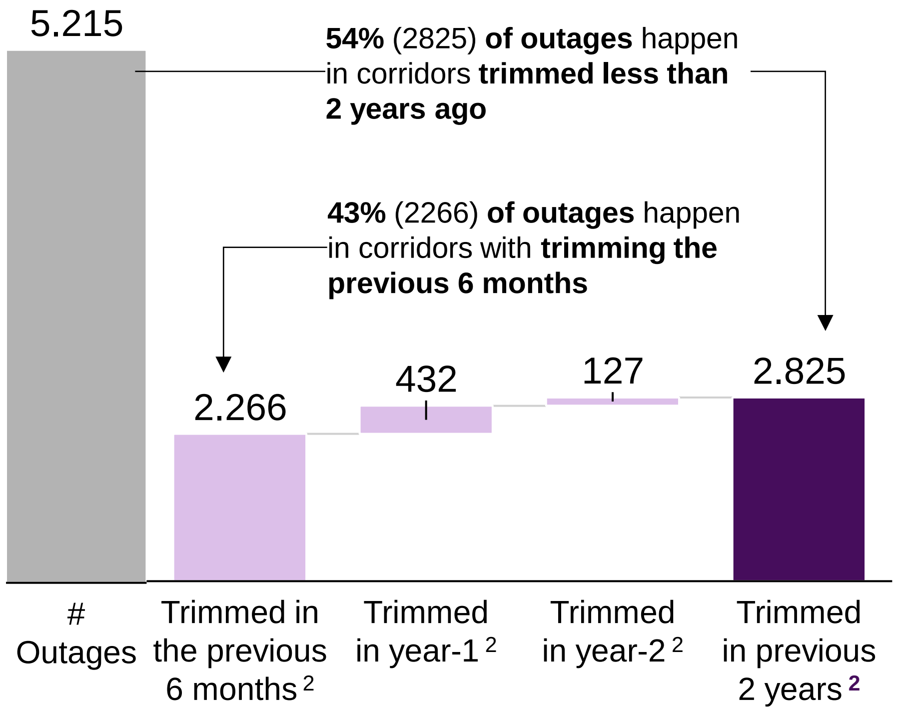
We matched **473 (66%) SAP trimming work orders to 1919 corridors** [1] by associating each work order to a set of devices in the trimming path

~50% of the corridors[1] with past trimming had an outage later, against ~19% of corridors without recorded trimming had an outage later

For outages in corridors with trimmings in the previous two years (2825 instances), the **outages occurred on average 122 days** (~4 months) **after the trimming activity**

… and in the first months after the trimming activity

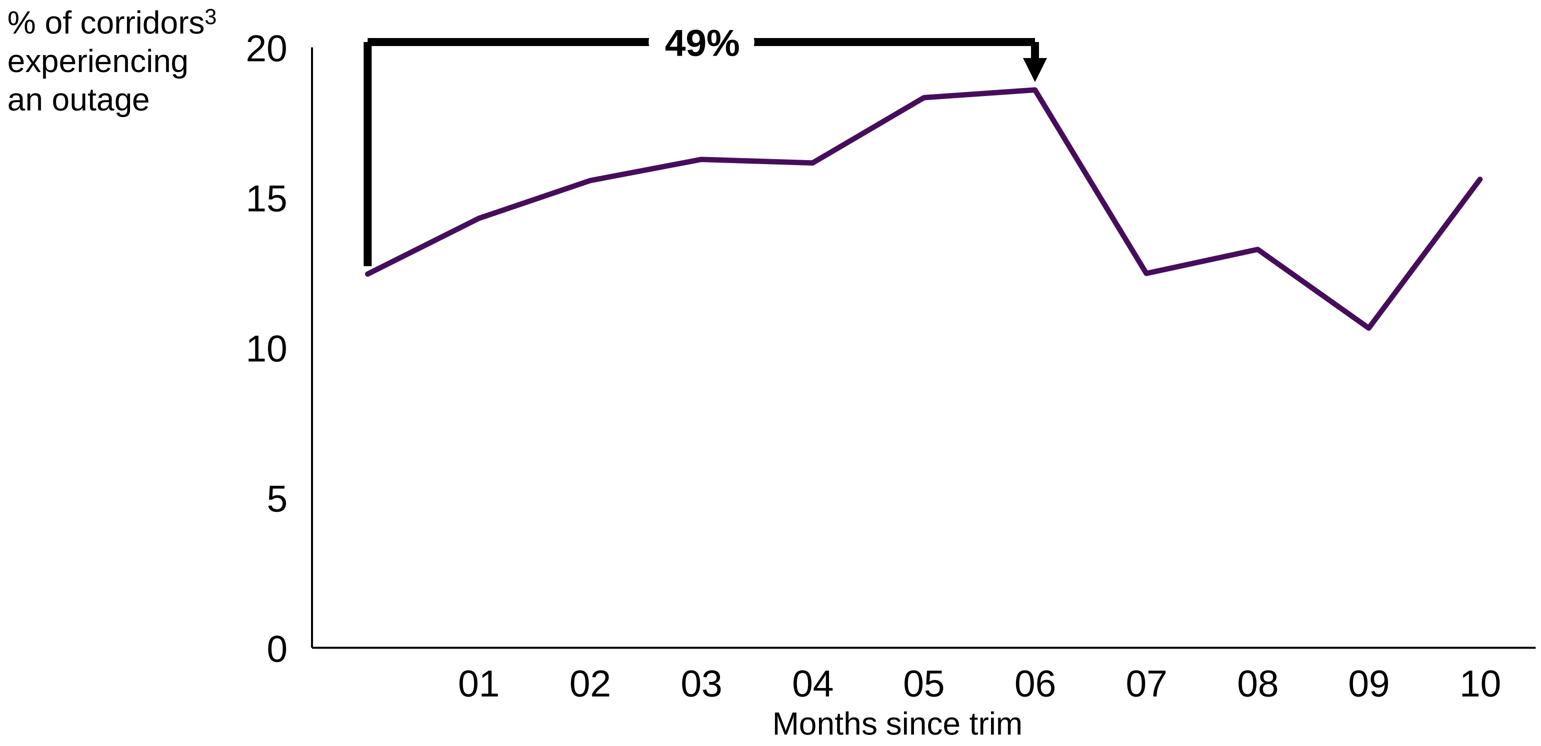
Distribution of trimming activities before outages



1 Corridor-Year pair; 2. At least one trimming point recorded in the corridor

2. Corridors trimmed frequently shows a fast increase of outage risk in the first 6 months

Percentage of corridors with outages vs Months since trim[1] for corridors trimmed frequently[2]



Outage percentage increases by approximately 49% (from 12% to 18%) **over the first six months** following a trim, **with an average monthly increase of 6%** during this period

Corridors trimmed frequently:

- **Are in high-risk areas**, as the risk of outage increases rapidly during the initial months following trimming
- **Have the highest risk of experiencing outages in the first months after being trimmed**, leading them to fail multiple times

1. Only months with least 200 corridors were considered, removing the data points for high values of months since trim as the considered corridors were trimmed frequently
2. At least 4 trimmings in the past 4 years, on average one one per year
3. Calculated as: # corridor-months since trim couples experiencing an outage / corridor-months since trim couples

We performed data transformations to overcome the limitations of data and developed recommendations for future actions

Requirements	Issues found	Proposed solution	Recommendation
Categorizable events description	**The outage cause descriptions in OMS data are unstructured,** making it impossible to perform a granular analysis of outage categories	Filtered outages due to vegetation based on a series of terms referring to vegetation, e.g. “tree”, “grow”, “branch”	Create a list of possible outage cause categories and **ensure each outage is reported with one of the predefined standard categories** from the list
Connectivity between data sources	**SAP and MJM records could not be connected**	Included only historical SAP data in the model	**For each MJM entry, add the corresponding SAP ID** (available starting next year). Perform cross-checks to verify that every SAP ID is reported in MJM data
Data completeness	**Only 6% of the extracted SAP data includes the trimming location** and there is no clear data extraction method that guarantees the inclusion of this information	Manually copy pasted location information from SAP system	**Ensure all relevant columns** (ID, date, location, description, cost) **are available** for data extraction
Clear data model	The MJM GIS layers contain various types of data and the **data model lacks clarity regarding the meaning of different columns**, e.g., does "Creation Date" refer to the survey date or the trimming date.	Aligned with Vegetation and GIS teams on the meaning of date columns	Data columns should be explicitly defined with unambiguous labels and, in the case of GIS layers, **accompanied by metadata that specify their exact meanings**
Precise location recording	In MJM, the number of surveyed trees is recorded at individual points, making it **impossible to determine the start and end of the surveyed sections** to which the number refers	Associated each point with its corresponding line segment and aggregate the number of trees from each point to the entire line segment	When recording survey or trimming activities, **document both the number of trees and the specific section** of the line segment they correspond to
Overhead line positioning	**GIS of overhead lines are offset from their actual location by about 4-8 meters**	Buffer spans out 10 meters on either side to capture presence of vegetation	**Manually correct position of lines** to their actual geographic location

PEA
PROVINCIAL ELECTRICITY AUTHORITY

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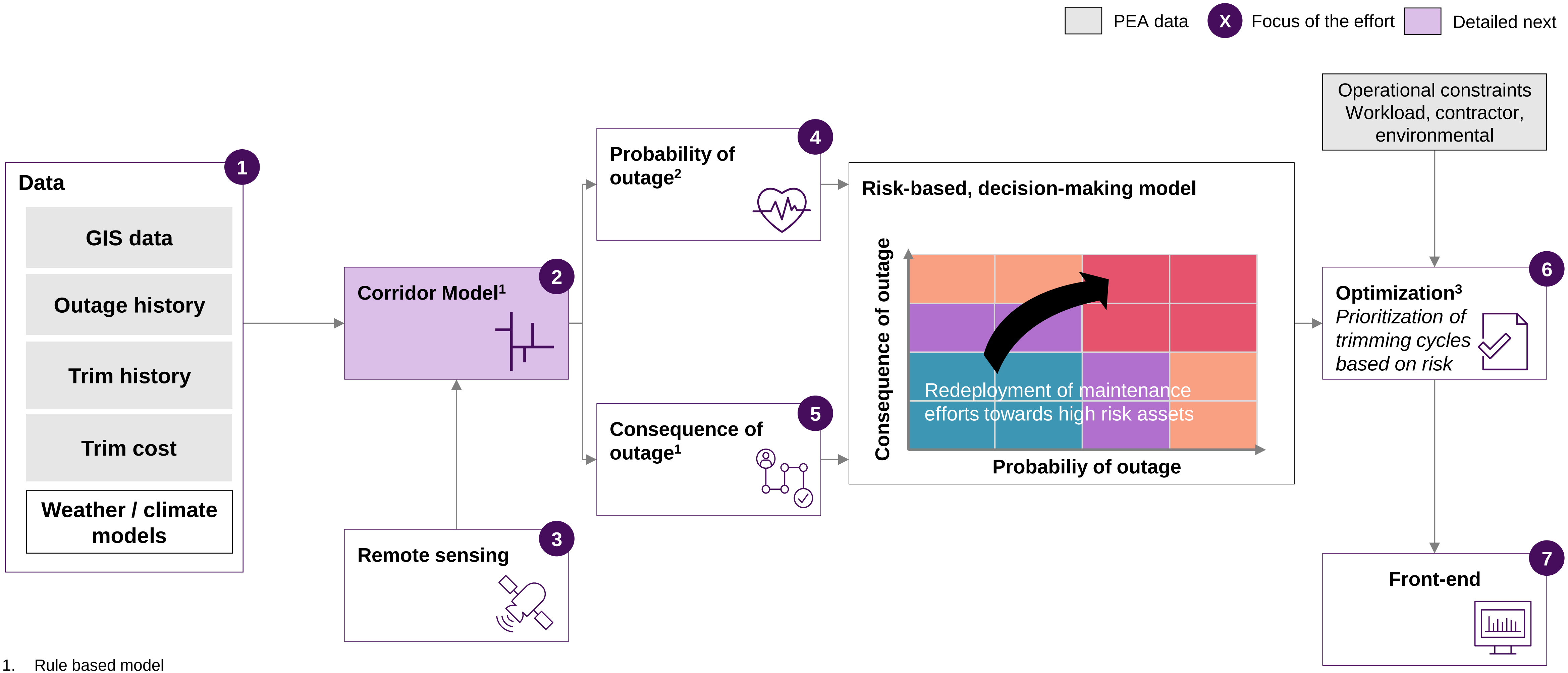


1. Rule based model
2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming

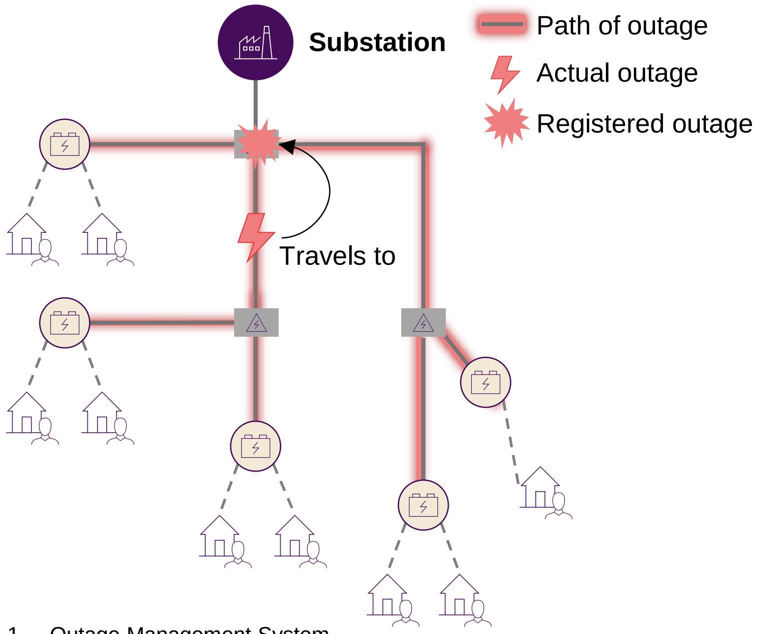
Source: McKinsey & Company

Our analytics approach begins with the extraction of “corridors” from GIS, to identify line sections

Protective device | Transformer | Primary Lines | Customer | Secondary Lines

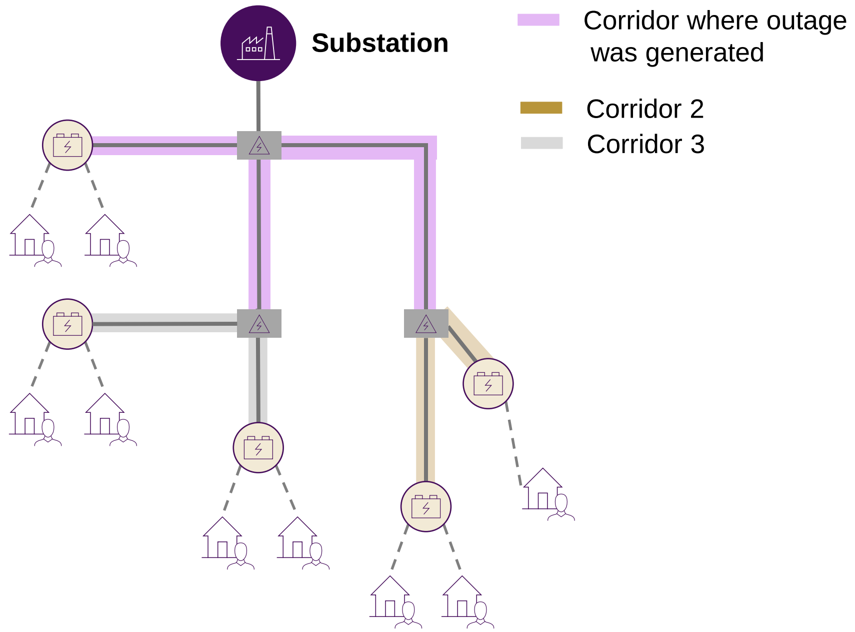
OMS[1] outage

If a line faults, the outage will travel upstream to the nearest protective device, and then all customers downstream of that device lose power.
Outages are assigned to the upstream protective device that records the issue.



Corridor mapping

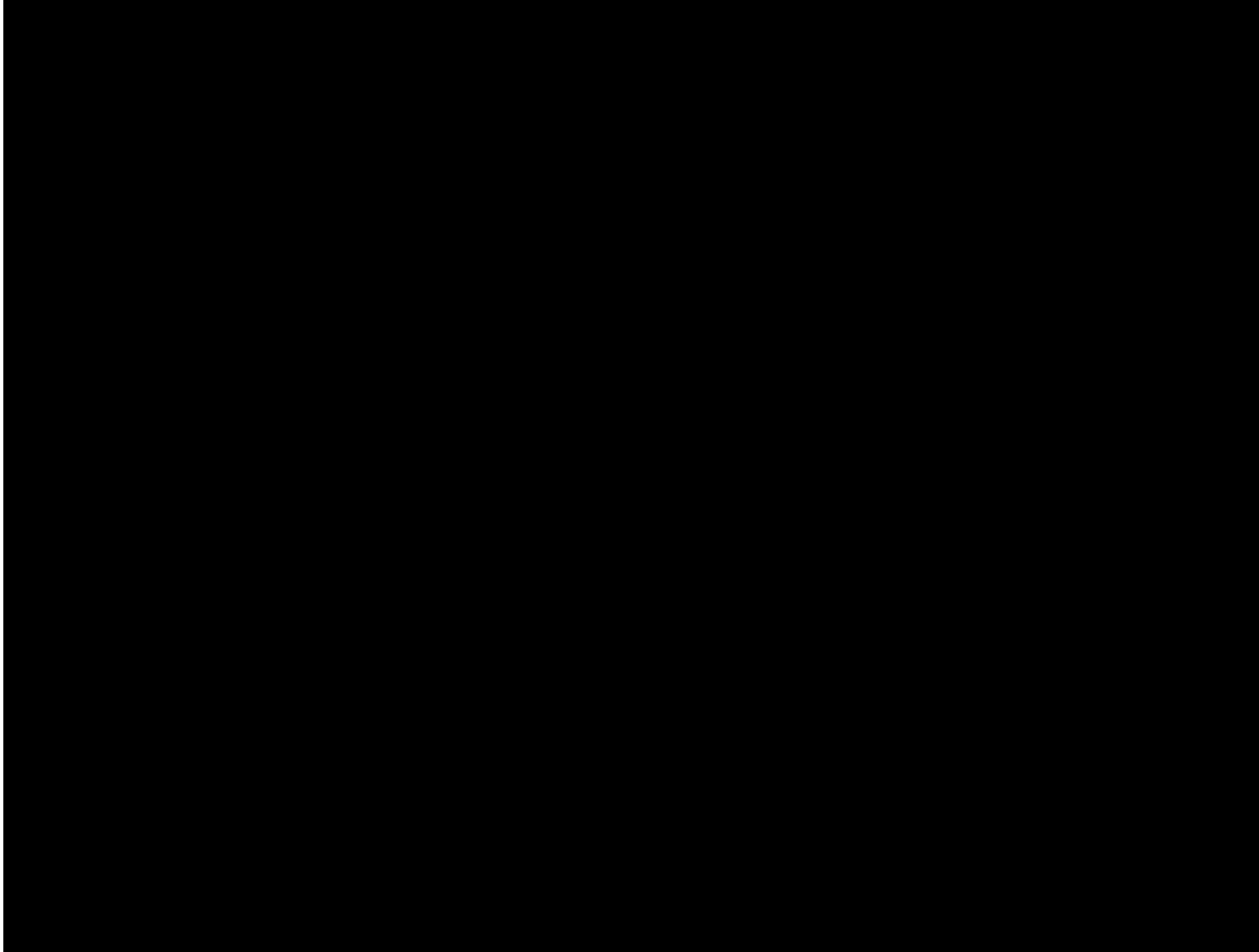
We **group lines connected to the same upstream protective device as a 'corridor**' to **attribute outages to the fewest lines**, forming the basis and lowest granularity level of our outage probability model



1. Outage Management System

- A **corridor** refers to the **group of overhead lines** that roll up to the same **nearest upstream protective device,** and therefore those that could have. **generated any outage assigned to it**
- Corridors represent the **lowest level of granularity** for **pinpointing where vegetation is causing problems** given how outages are assigned in OMS
- Our corridor mapping tools operate similarly to the results of GIS trace tools, but **can be applied at scale to every overhead span in system automatically** and simulate changes in switching.

Corridor model in Phunphin



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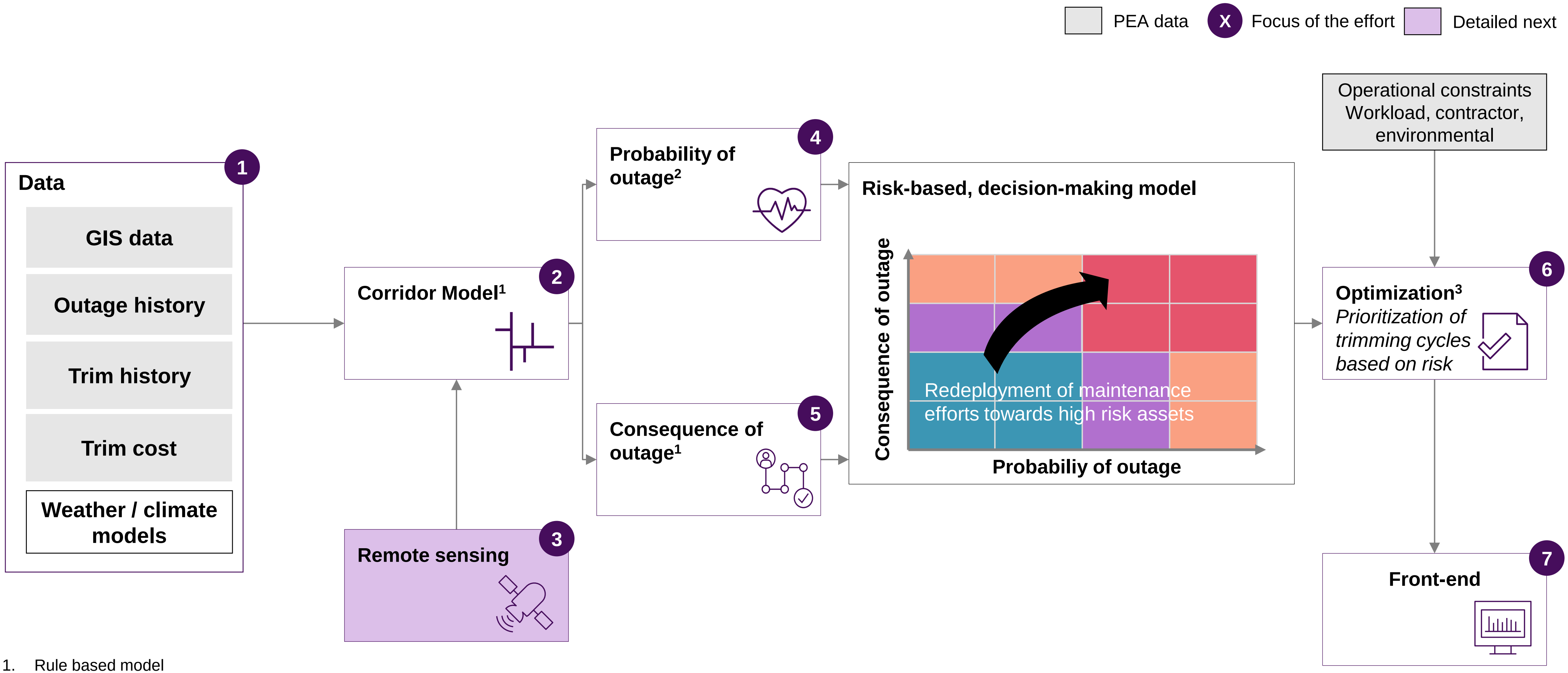
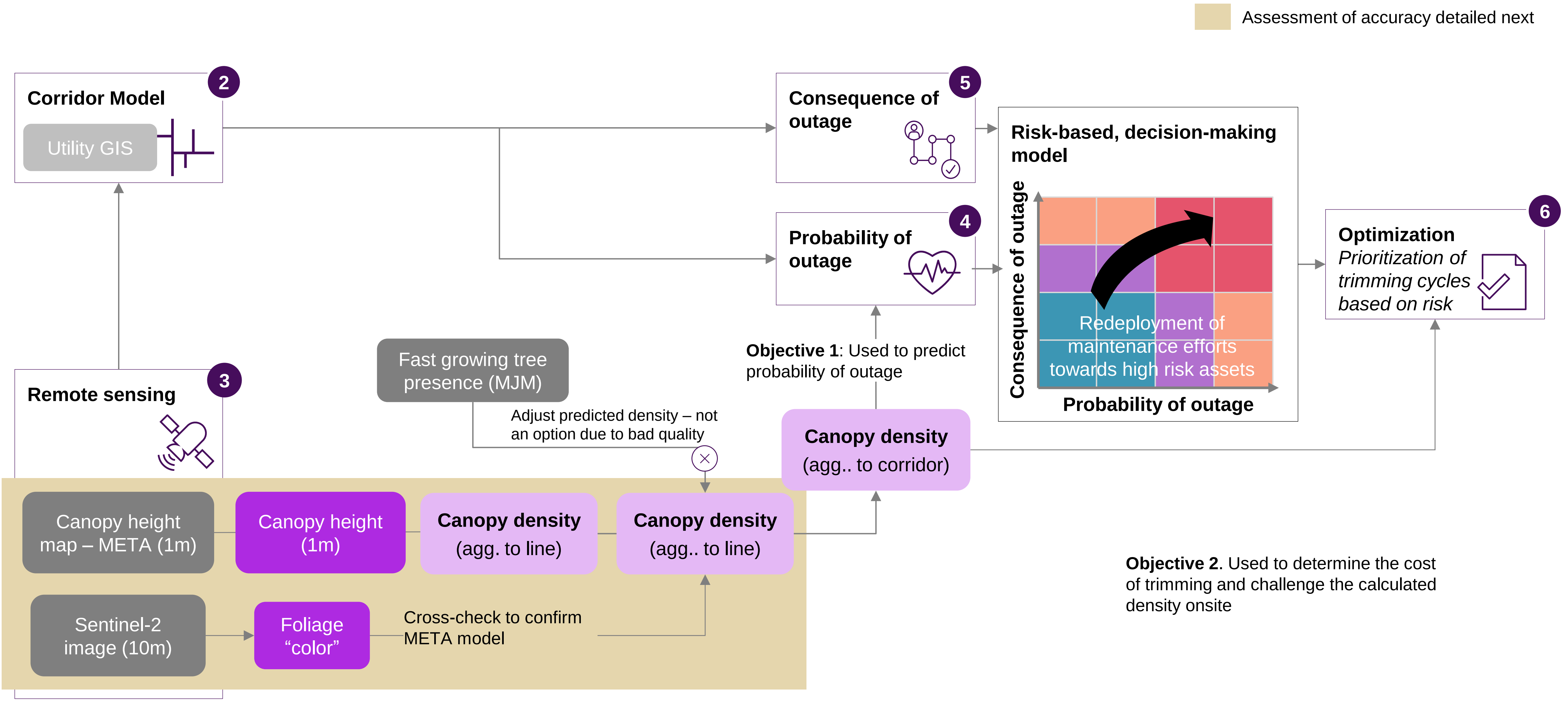


1. Rule based model
2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming

Source: McKinsey & Company

Remote sensing data provides explicative variables used for the outage model and the estimation of the trimming costs

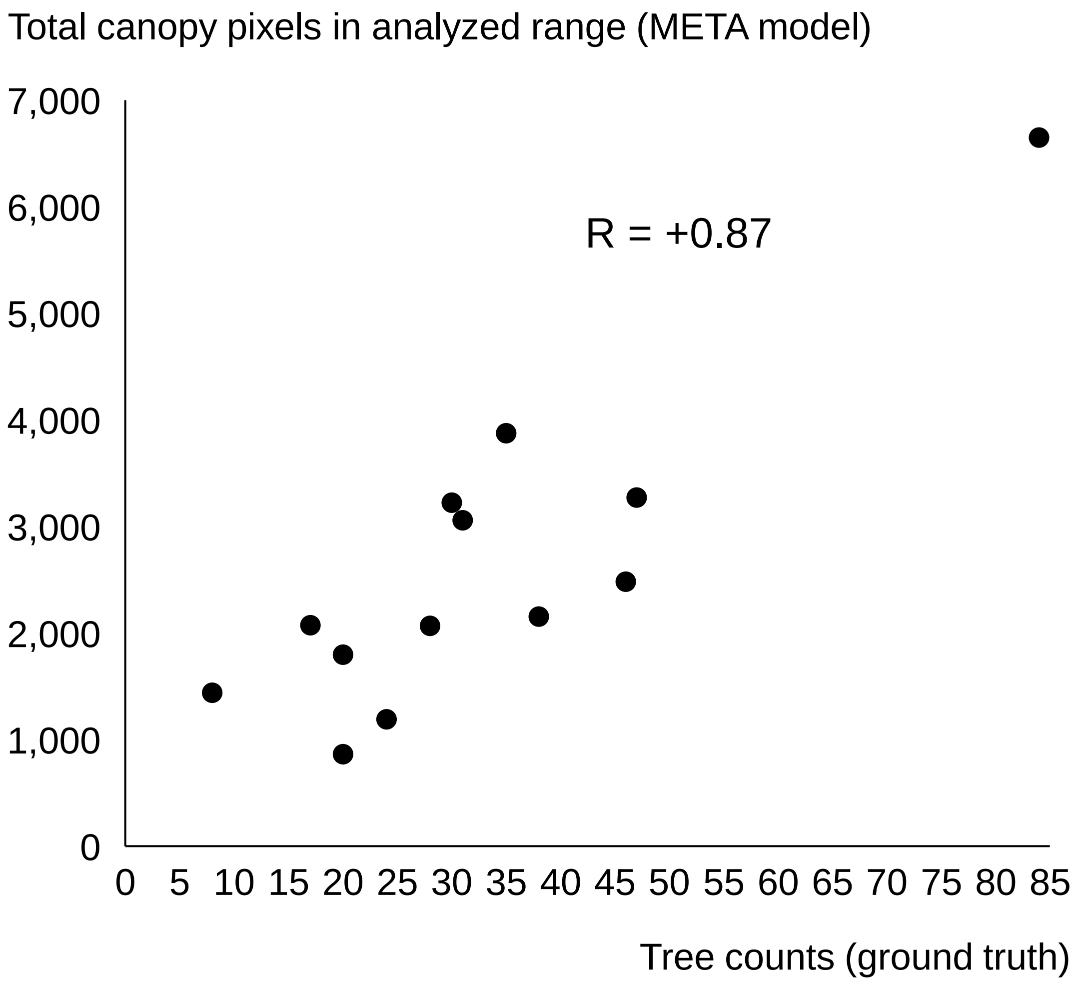
Remote Sensing data processing pipeline



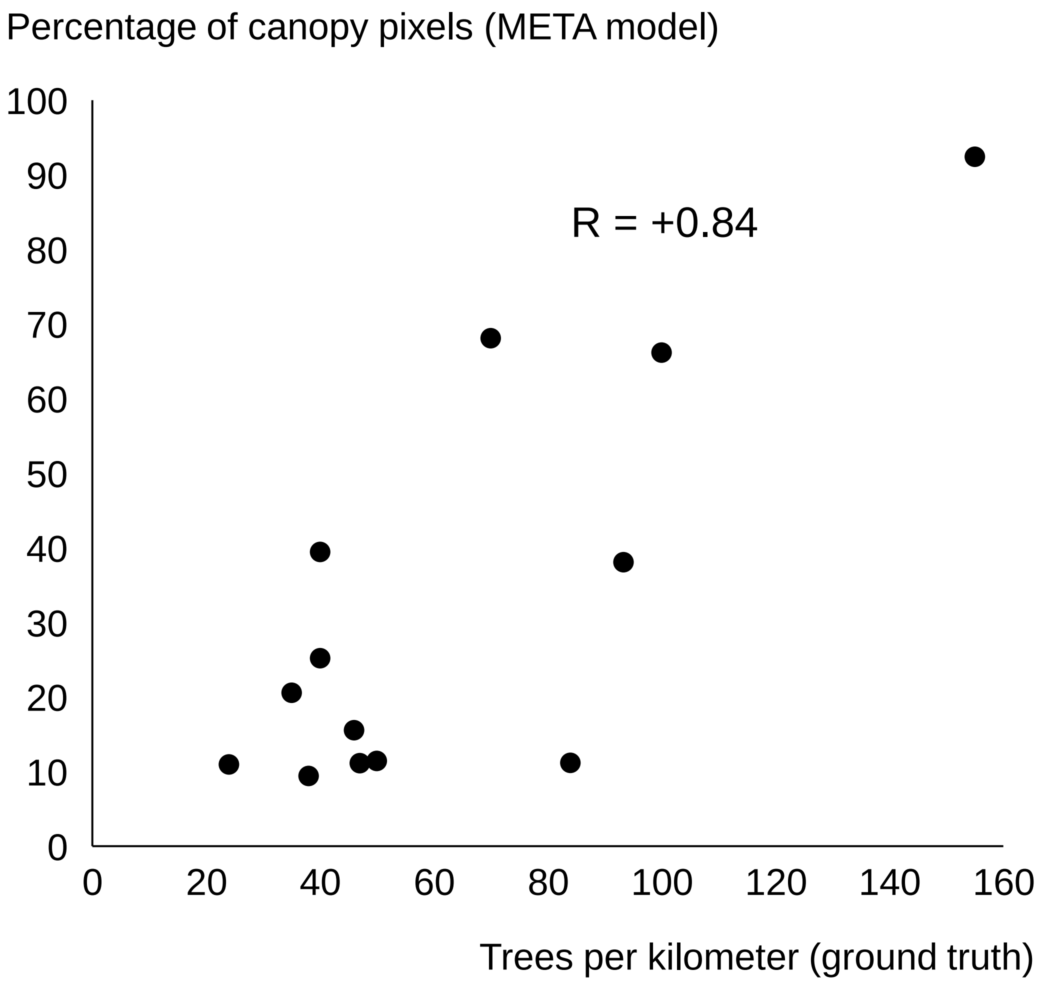
Source: McKinsey & Company

The META model achieved a strong correlation with field observed density which we will seek to match or exceed with Sentinel-2

META model
Total canopy pixels vs real tree counts

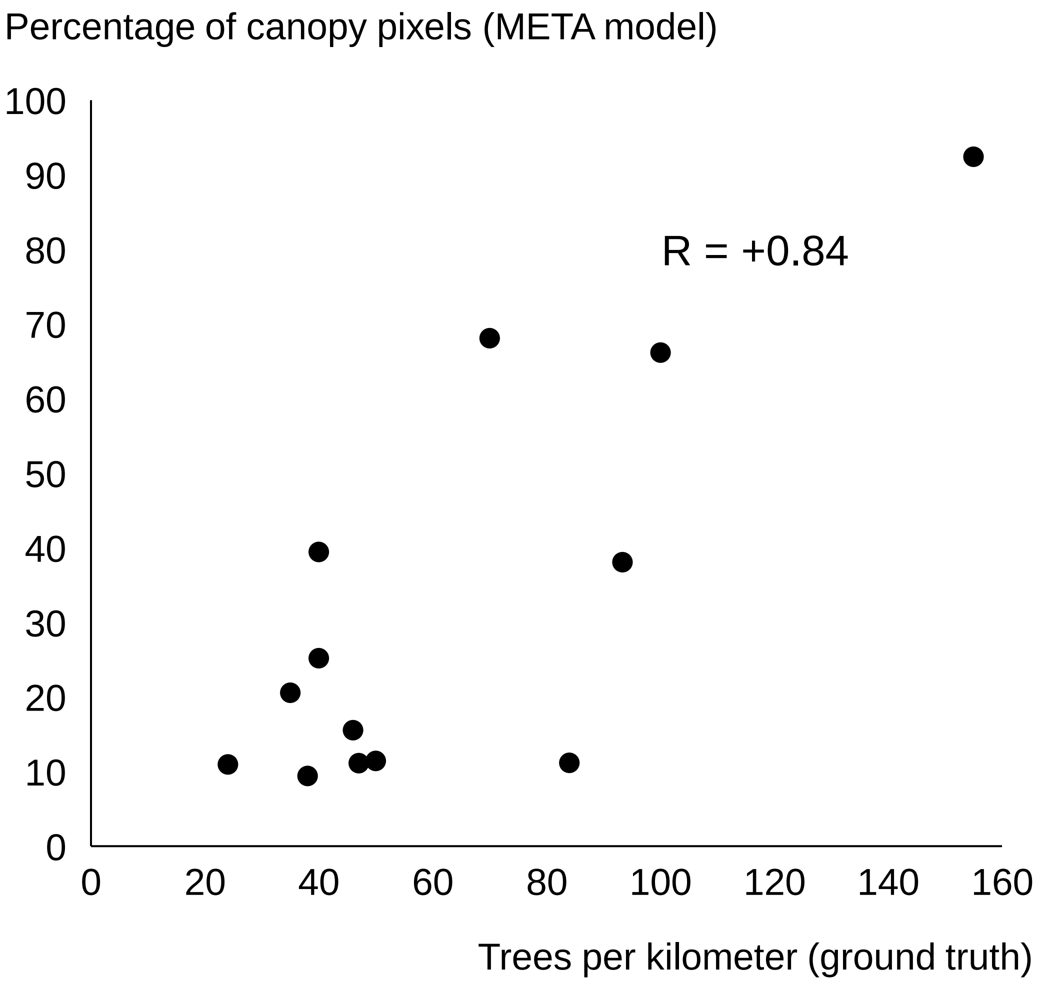


META model
Normalized density vs trees per km

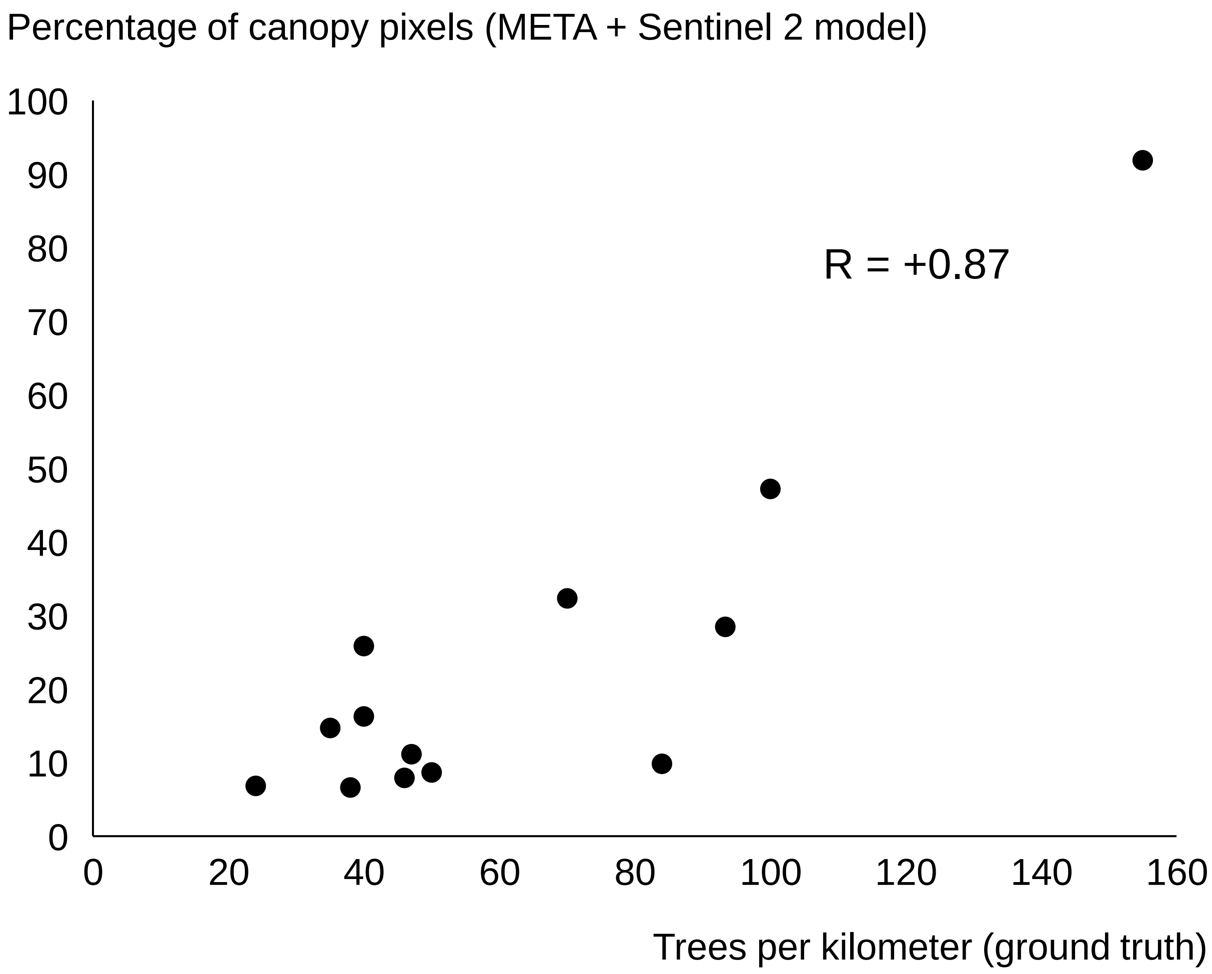


Adding Sentinel 2 to the model has increased the correlation by three points

META model
Normalized density vs trees per km



META + Sentinel 2 model
Normalized density vs trees per km (2024)



PEA
PROVINCIAL ELECTRICITY AUTHORITY

The current methodology for combining META and Sentinel 2 applies the rate of density change from the years META was trained to current year

Methodology for density calculation leveraging Meta + Sentinel 2

	META	Sentinel-2			META + Sentinel 2
Line	**Baseline density (2017-2020)**	**Average vegetation density (2017 – 2020)** –	**Vegetation density in current year (2024)** =	**Change in vegetation density (2017-2020 to 2024)**	**Adjusted vegetation density to 2024**
1	31%	40%	35%	-5%	29%
2	20%	21%	30%	+9%	22%
3	50%	35%	41%	+6%	53%
4	10%	14%	11%	-3%	10%
5	15%	11%	16%	5%	16%
		*Average of sentinel-2 foliage pixels for **same years as META was collected***	***Current year** sentinel-2 foliage pixels*	***Change from META years to current year***	*Baseline density * (1+ change in vegetation density)*

31% + 31%*(-5%) = 29%*

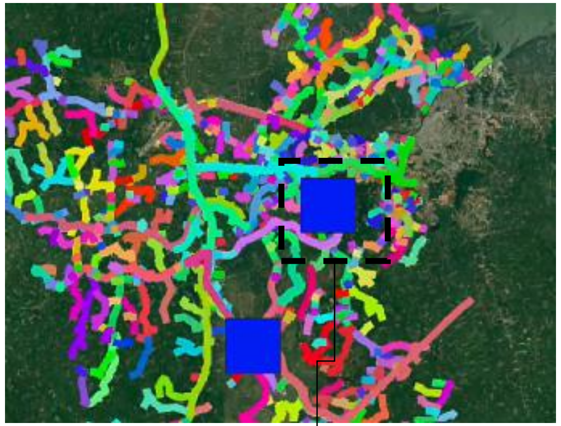
Remote sensing variables extracted will add predictive power to probability of failure model, as well as determine density for trimming cost determination

Demo video

Remote sensing variables are extracted for corridors, the lowest level of granularity in lines, to predict vegetation-related outages and determine trimming cost automatically

Lines are split into the lowest level of granularity: corridors

Substation | Protective device | Line, colored by corridor ID



A **corridor** is defined as the **group of overhead lines** that roll up to the same **nearest upstream protective device**

Given GIS data, our corridor mapping solution **automatically splits lines into corridors**, lowest level of **granularity** for:

- **Risk calculation**
- **Trimming optimization**

Remote sensing variables are extracted for each corridor to help predict likelihood of vegetation-related outages and determine density for trimming cost determination

1 Baseline canopy density in 2020 – META model[1] (1m)

1m resolution

META model trained on 2018-2020 data

2 Density change from 2020 to 2024 – Sentinel – 2 (10m)

10m resolution

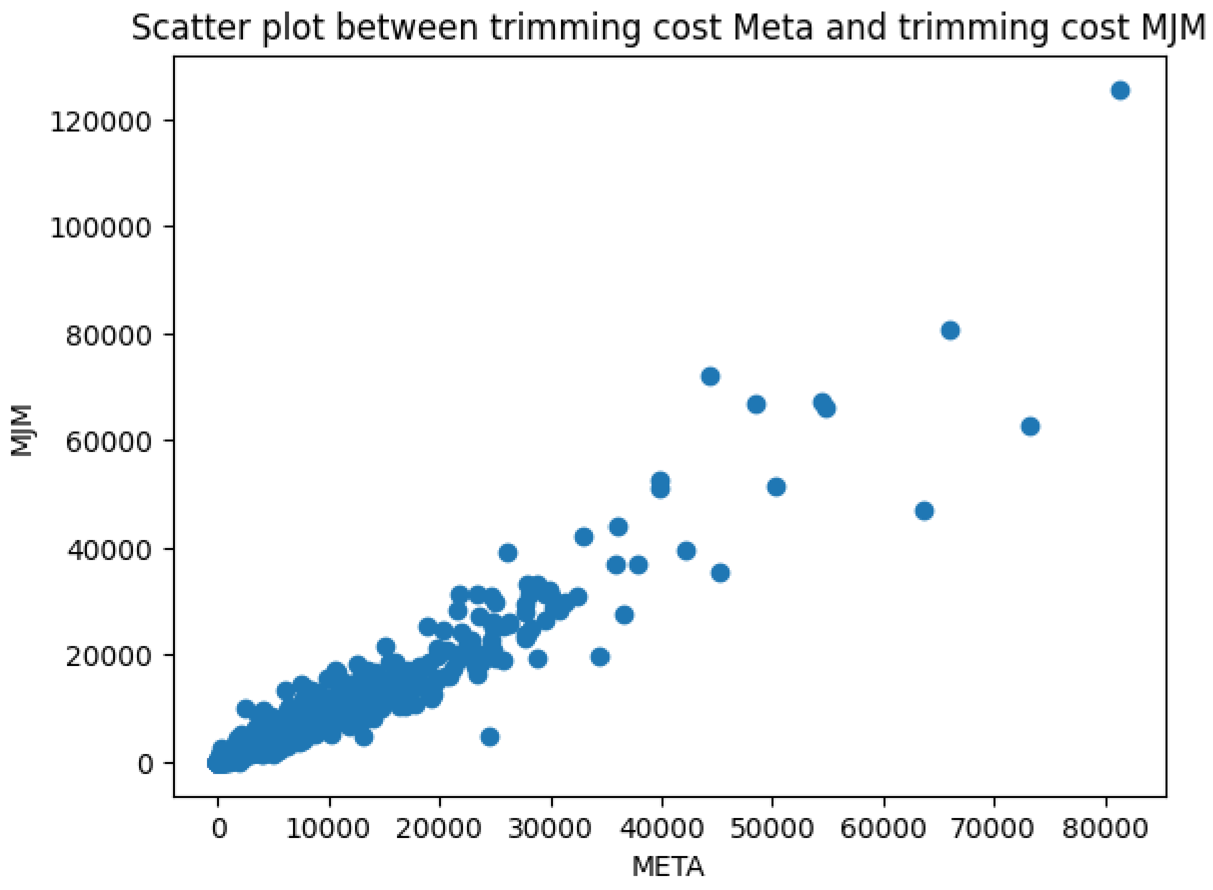
Sentinel -2

3 Density estimation in 2024 – META + Sentinel-2 (1m)

1. META model trained in 2018-2020 data

META and MJM-based trimming costs correlate at corridor level

Trimming cost of corridors as sum of line trimming costs calculated based on the density bin associated with META and MJM



$R2 = 0.89$

Pearson = 0.95

Mean Absolute Error: 1130

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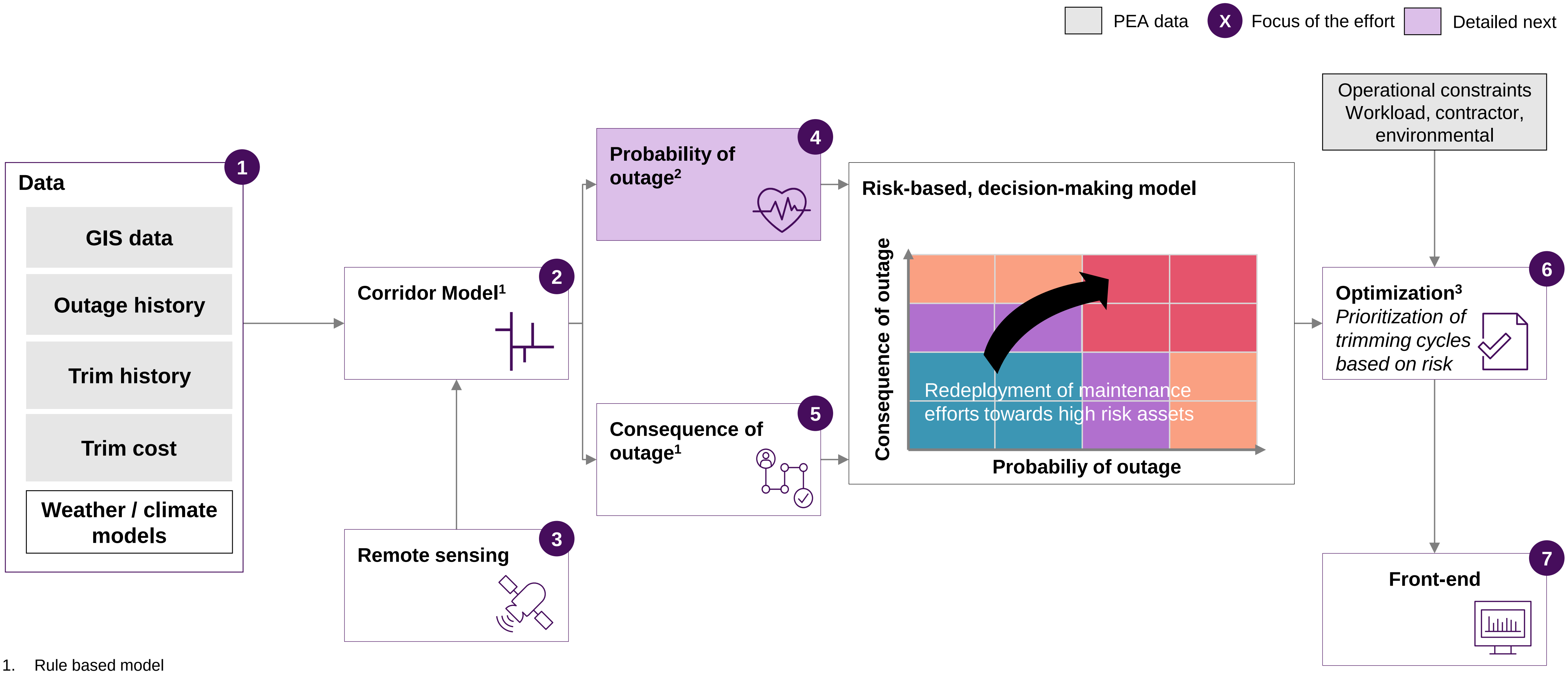

1. Rule based model
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3. Mixed Integer Programming

Source: McKinsey & Company

PEA PROVINCIAL ELECTRICITY AUTHORITY

The model is co-created with PEA in an iterative way following our Advanced Analytics methodology

Analytics model creation follows a 7-step iterative cycle, with results tested and refined with SME[1]s

X Detailed next

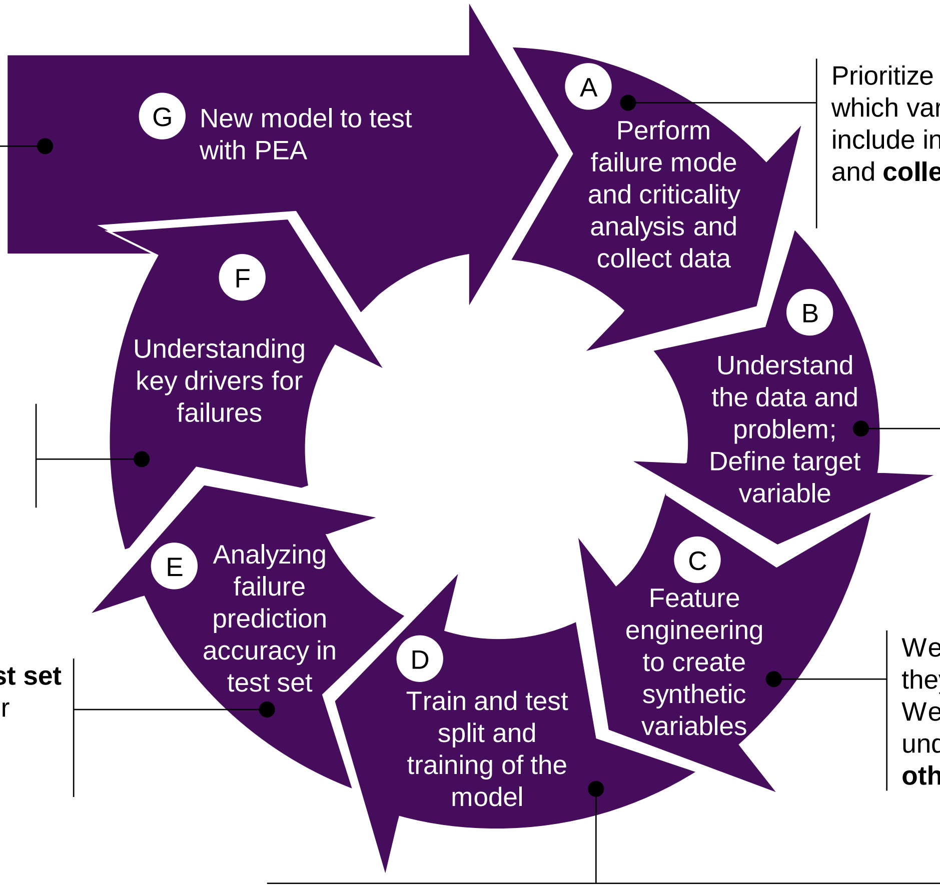
Our methodology for agilely creating analytics models is based on a 7-phased iterative cycle

Results are generated to test with the PEA and refined during the next cycle

PEA feedback is collected to incorporated it into the next modeling cycle

We analyze features driving the prediction, their **directionalities and importance** and obtain actionable insights

Model's predictive power in test set is measured via decile analysis or GINI metrics



Prioritize failure modes to help indicate which variables are most important to include in the new maintenance plan and **collect the corresponding data**

We discuss the problem and data limitations; We **define which of the given variables are considered the target** of the prediction according to the nature of the problem

We **add all the data sources in a master table** where they are consolidated and related to each other
We **deal with the variables** incorporating the understanding of PEA experts to **add new or eliminate other correlated** without physical sense

Data is split into train and test: **train subset** is leveraged by the model to **learn patterns between input data and target** (failures), while test set is used to assess model's accuracy

1. SME: Subject Matter Expert

A. GIS, outages, trimming data and external data will be leveraged to develop a outage probability model

PEA data | External data

4 years of information | **5215** Outages | **711** SAP trimmings | **1646** assets | **3008** Km

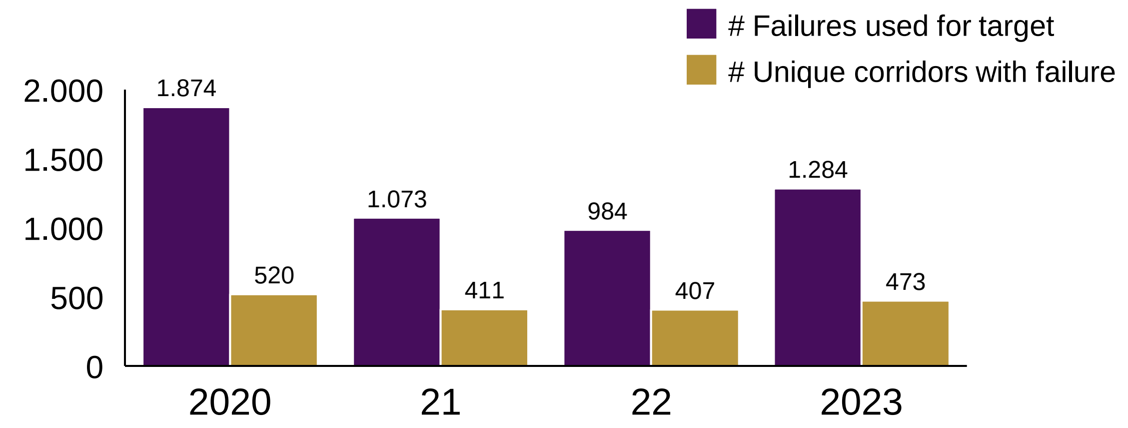
Data source	Variables and descriptions	
GIS data (Layers: circuit breaker, switch, lines, transformer, recloser, poles)	• **Asset:** Corridor - Group of MV overhead lines rolling up to the same upstream device • **Type of cable:** aerial, aluminum conductor steel, partially insulated etc • **Lat and Lon:** Latitude and longitude of the asset	• **Customer number:** Number of customer per transformer • **Customer type:** Number of customer per area (1,2,3,4,5) per feeder • **Length**: Length of the corridor
Outages (OMS system)	• **Asset:** *OpDeviceGIStag* column is used to identify the device affected • **Type:** Outage status	• **Outage Date Time:** Time of outage occurrence • **Cause:** Description of the outage cause, filtered to vegetation-related
Trimming data (SAP and MJM systems)	• **Asset:** Section or point on the line • **Date:** Date of trimming activity	• **Order:** Unique ID of the trimming activity • **Km trimmed:** Total km trimmed for each work order • **Fast growing**: Presence of fast-growing vegetation
Weather data (Meteostat[1] package)	• **Temperature:** Daily station measurement • **Precipitation:** Daily station measurement	• **Wind speed:** Daily station measurement • **Atmospheric Pressure:** Daily station measurement
Remote Sensing data	• **Canopy density:** Density of trees in given radius (e.g., 10m)	• **Canopy height:** Height of vegetation in given radius (e.g., 10m)

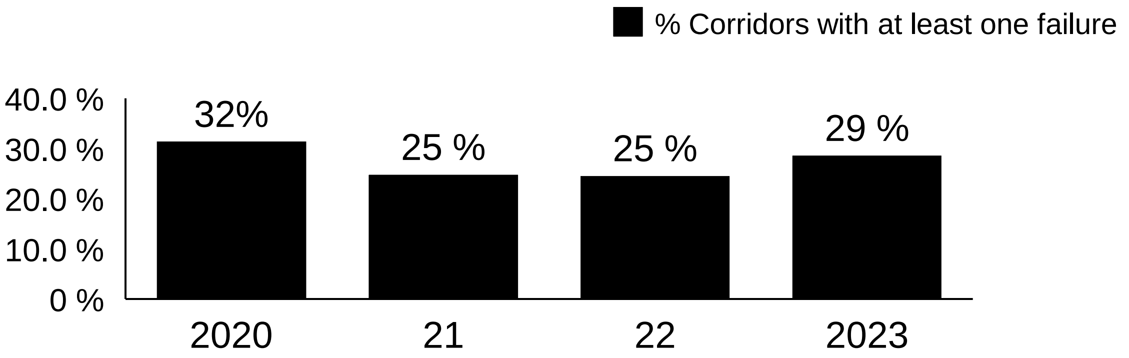
1. Meteostat - public source with meteorologic data at the weather station level; 3. About 53% of age information are still missing so the avg age will still change

B. OMS Outages have been defined as the target variable that identifies unhealthy corridors each year

Vegetation-related failures[1] over time

- Based on data availability and model capturability, a **target for each pair (corridor, year) has been defined**
- A combination (corridor - year) is said to be anomalous **if the upstream device of the corridor has had a vegetation-related failure in the given year.** Leveraging failures as target event is a common practice on utilities





- With this target definition, an **average of 27% of all corridors** (1646) are labeled as anomalous each year
- On average, **faulty corridors fail 3.6 times** per year

1 Filtering out 4% of outages due to vegetation during storms

C. 71 features have been created to test in the preliminary health models to help predict failures

Full list of created features here

Included | Partially included | Future iterations | (X) Created features per source

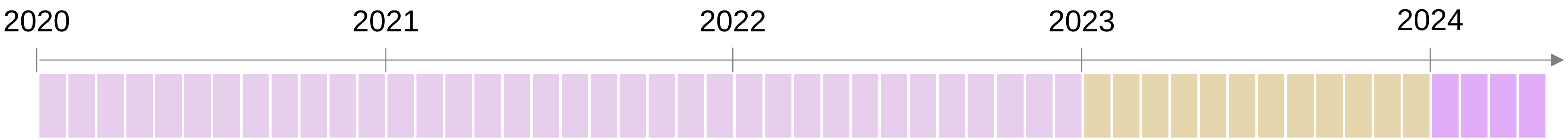
Source	Raw variable	Transformation description	Tr. timeframe	Example of feature created
GIS data (7)	Cable material	Categorical variables are transformed into one column per category and then the categories are summed by asset		# lines with cables in bare aluminium
	Number of customers by transformer	Calculation of affected downstream customers		# downstream customers
Outages (12)	Outage date time	Binary column indicating failure in given year		To be used as target variable based on outage year
		Lag by asset	Last 1-3 years	# outages in the past year
	Cause (e.g., tree growth, branch falling)	Transformed into one column per cause and lagged	Last 1-3 years	# tree grow-in caused outages in the past year
Trimming data (16)	Order	Cumulative number by asset		Cumulative # of trimmings in past 3 years
	Date	Difference with respect to current date Season derivation		Years since trimming # trimmings in dry season
	Km trimmed			Km trimmed in the same Order
Weather data (31)	Temperature, wind speed (daily)	Avg, max, min by location and season	Last year	Average temperature last year
	Precipitation (daily)	Max and sum by location and season	Last year	Sum of precipitation last year in rain season
Remote Sensing data (5)	Canopy height in 2m pixel	Calculate the canopy presence as % of vegetation pixel in 10m buffer around the line. Max, sum, km-weighted mean of lines in the corridor		Mean density in corridor, max canopy height in corridor

D. The outage probability prediction will be performed by a machine learning model trained with 2020-2022 data and tested with 2023 data

ILLUSTRATIVE

Train | **Test** (detailed next) | **Implement**

Train / Test split for the failure prediction model



	Train	Test	Implement
Subset	**2020 to 2022** (3 years, 75% of data)	**2023** (1 year, 25% of data)	**Future months**
Objective	**Train the model and learn patterns between the input** variables **and target** response (outages)	**Test the model** by predicting the probability of failure leveraging the input data. **Predictions are then compared to real failures** never seen by the model	Production-ready heath models could be used to **predict the likelihood of failures to prioritize maintenance** in future months
# Failures	**1338**	**473**	

Split is done ensuring same target distribution

Train/test split can be done either by assets or years; years is preferred in the context of a forward-looking view of maintenance planning and yearly spend

E. The model learns relationships between input and output in the training set, and is then tested on unseen data to verify the accuracy of these rules

Training and testing process of outage probability model for corridors

Train | **Test** (detailed next)

Input - Train **INPUT** **OUTPUT** **Output - Train**

Nearest upstream device	year	# lateral lines	# insulated lines	Average pole height	Sum precipitation	Tree-fall outages year-1	Years since trim	Outage
4284SW000000582	2020	15	11	12.2		0		1
4284SW000000582	2021	15	11	12.2	2148.9	3	1	1
4284SW000000582	2022	15	11	12.2	2092.8	1	2	1
42SWKA000094791	2020	22	14	12.2		1		0
42SWKA000094791	2021	22	14	12.17948718	2345.3	0	3	0
42SWKA000094791	2022	22	14	12.17948718	2148.9	0	1	0
4284SW000000582	2023	15	11	12.17948718	2092.8	0	1	0
42SWKA000094791	2023	22	14	12.17948718	2345.3	0	2	1

Input - Test **Output - Test**

75% of data is used to train the model and **generate rules/formula (f)** between input and output:

f(*INPUT_train*) = *OUTPUT_train*

25% of data is used to **test whether** the generated **formula works in data not seen by the model** (test) and is able to capture failures:

1. We generate predictions by applying the learnt rules: **f(*INPUT_test*) = *Prediction***
2. We **compare the *Prediction*** to the **actual *OUTPUT_test***

D. Machine Learning model takes as input all created features and automatically selects only those related to the output (outage) to generate rules that correlate both and will help us predict outages in the future

Training process by Machine Learning algorithms – example for Corridors

Data leveraged by model

INPUT Train

71 explanatory features

Feature 1
Feature 2
Feature 3
Feature 4
Feature 5
Feature 6
Feature 7
….
Feature 69
Feature 70
Feature 71

OUTPUT Train

Failure: Outage related to vegetation

Model training

Model learns **correlations between INPUT and OUTPUT**

Assigns an importance to each feature, depending on how good this feature **differentiates between outage and non outage**

	Importance
Feature 1	**10%**
Feature 2	**3%**
Feature 3	0%
Feature 4	**4%**
Feature 5	**2%**
Feature 6	0%
Feature 7	**4%**
….	….
Feature 69	0%
Feature 70	**12%**
Feature 71	0%

Model building

Creates rules (f) with only relevant features

(59 features for vegetation)

f(INPUT_train – 59 features) = OUTPUT_train

Ignores features with 0% importance

(remaining 11 features)

Model testing

1. **Applies same rules (f) to relevant features in INPUT_test**

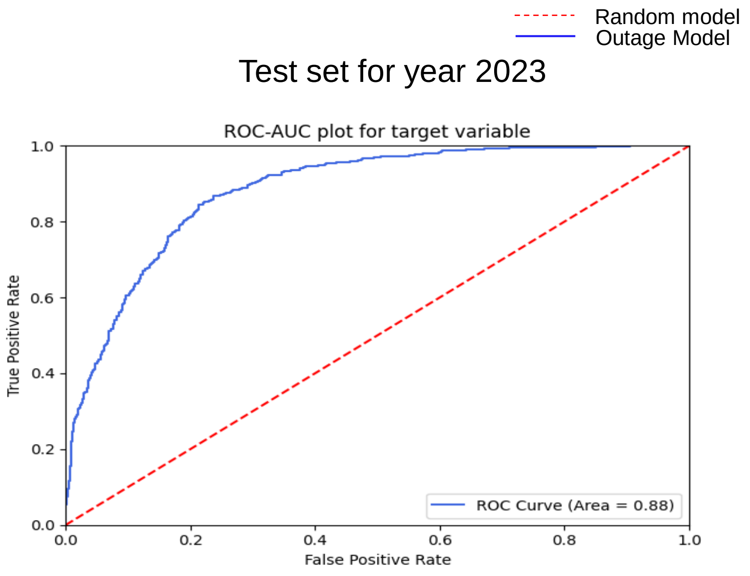
f(INPUT_test – 59 features) = predicted probability of outage

2. **Compares predictions to real outages**

Prediction	OUTPUT Test
20%	1 (outage)
2%	0 (No outage)
0,5%	0
40%	1
…	…

E. The initial health model achieves a GINI of 78...

Model performance, GINI



Name	Gini
Outage probability model	78

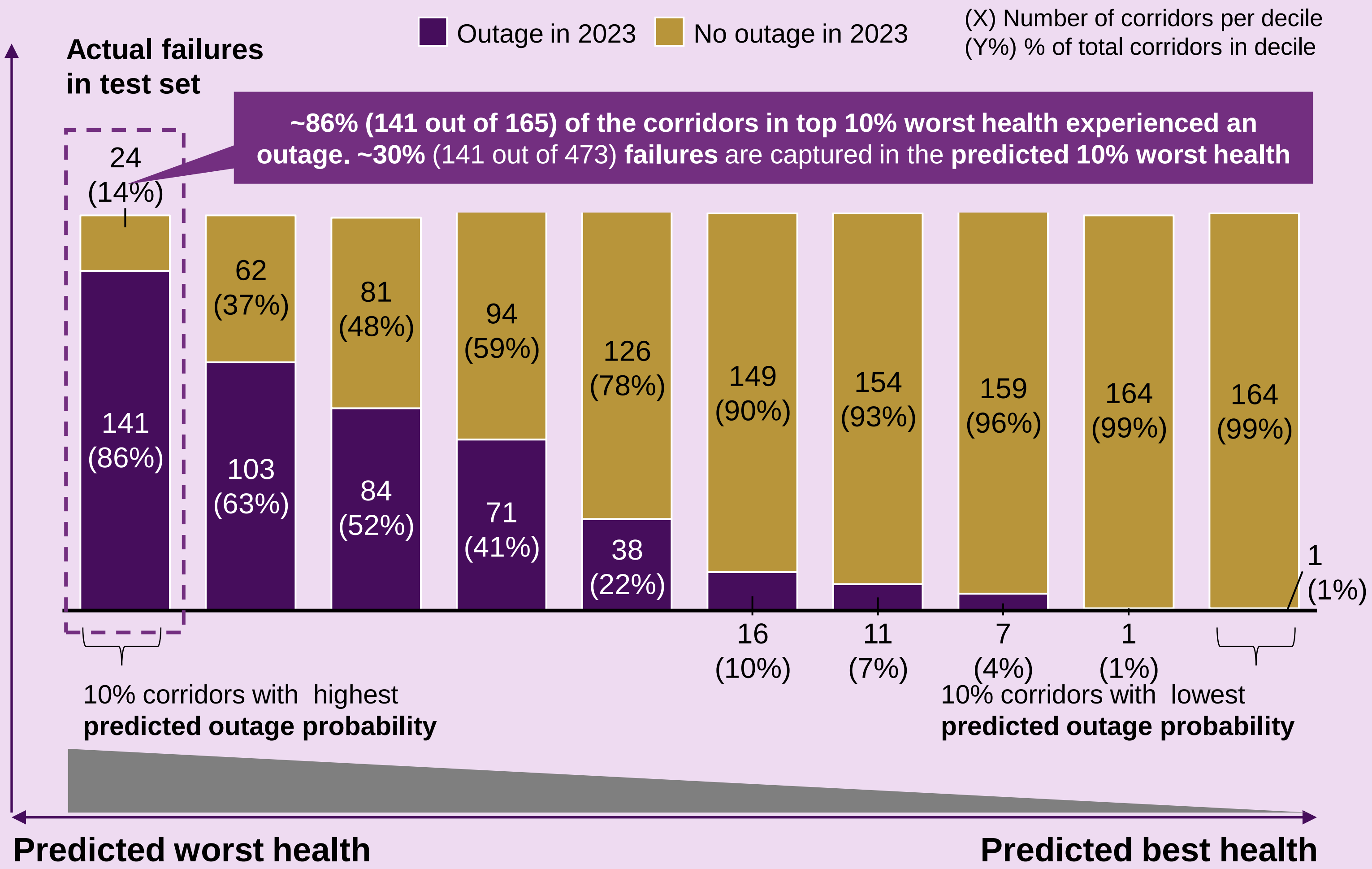
A GINI coefficient is a score of predictive effectiveness – it measures how well a model can differentiate between different classes or outcomes (0-100)
Typical engineering models obtain GINIs around 30, so we use the given value as a minimum threshold to go ahead

…and ~86% of corridors in in the worst 10% of predicted health result in a vegetated-related outages

Results in test set for 2023

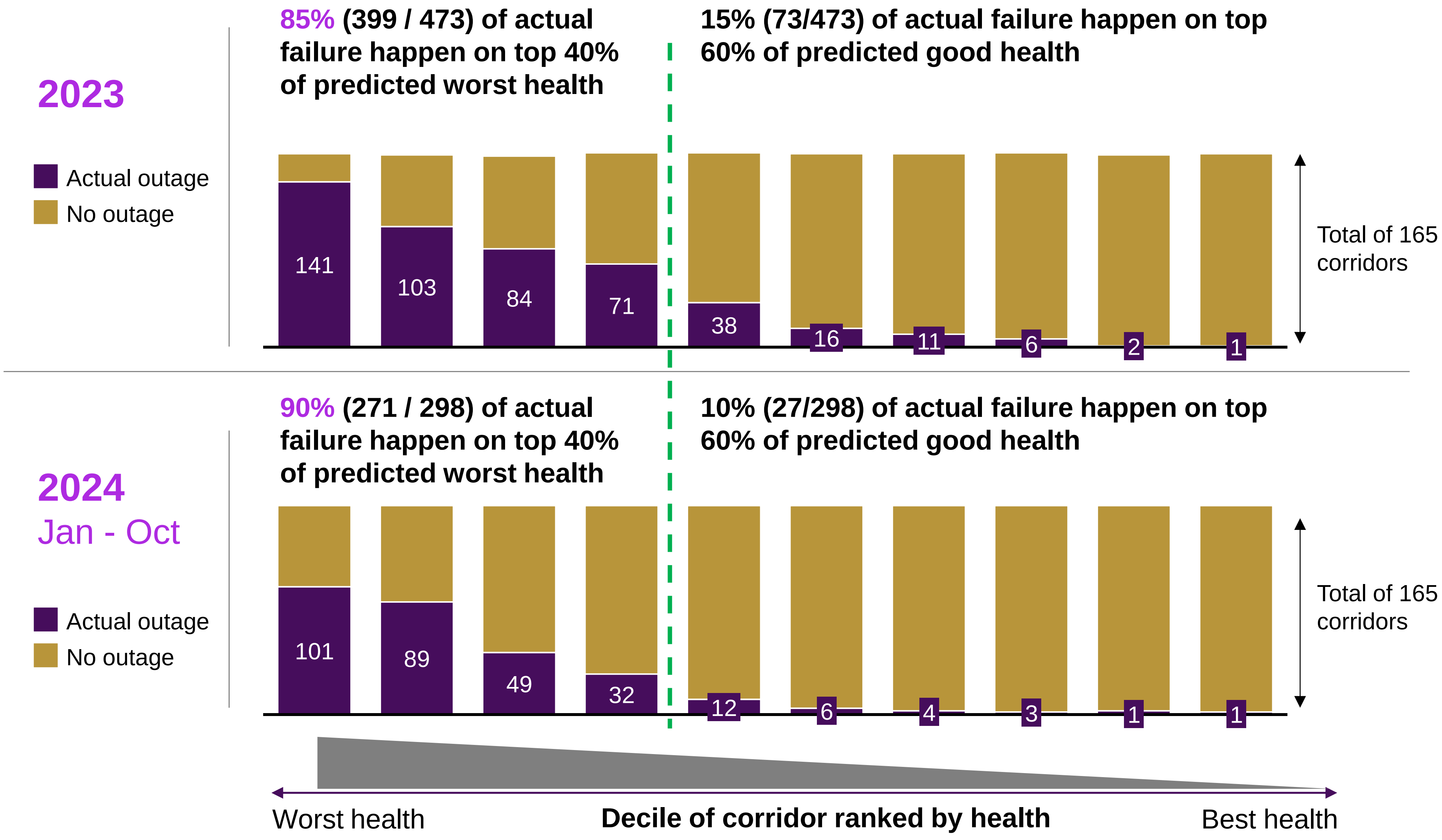
PRELIMINARY

Distribution of outages vs predictions for corridors in test set



E. Probability of outage: 85~90% of outages can be captured by top 40% worst health predicted by the model

Decile analysis

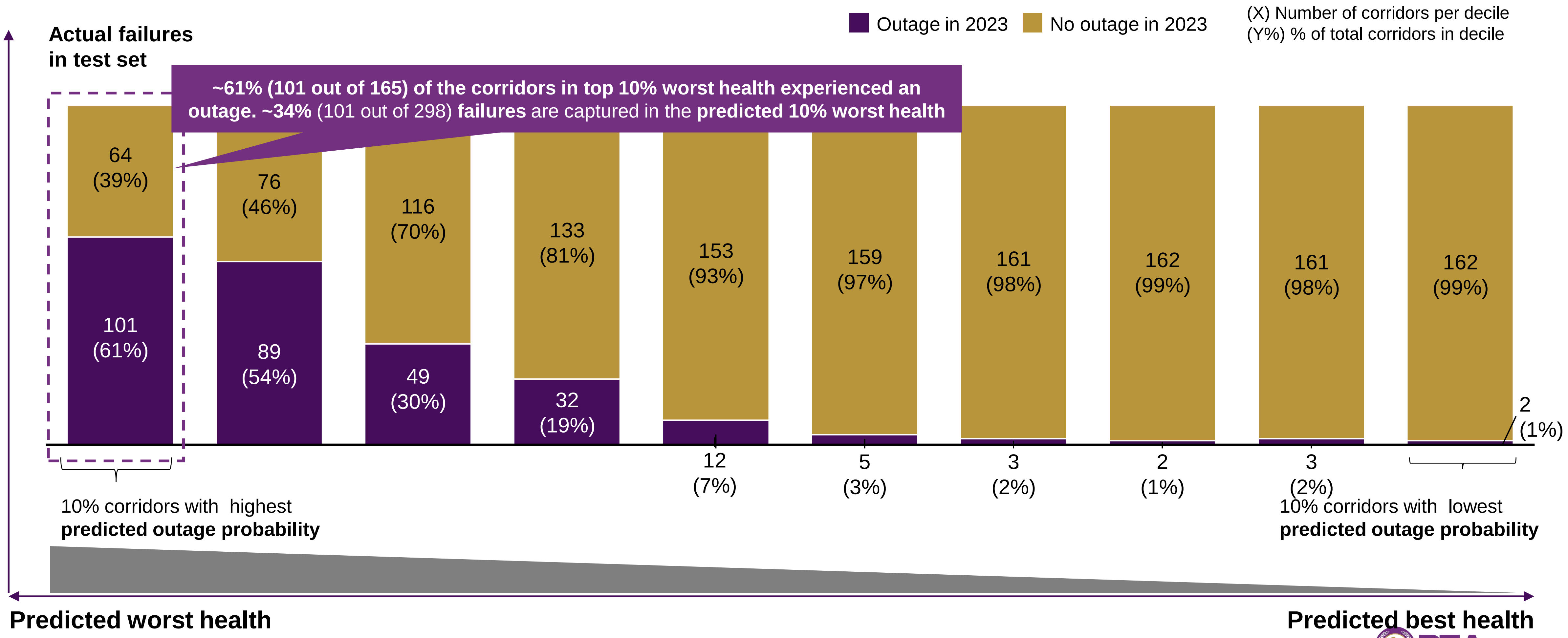


Probability of outages model perform well when comparing to actual outages

85 – 90% of outages can be captured by top 40% worst health

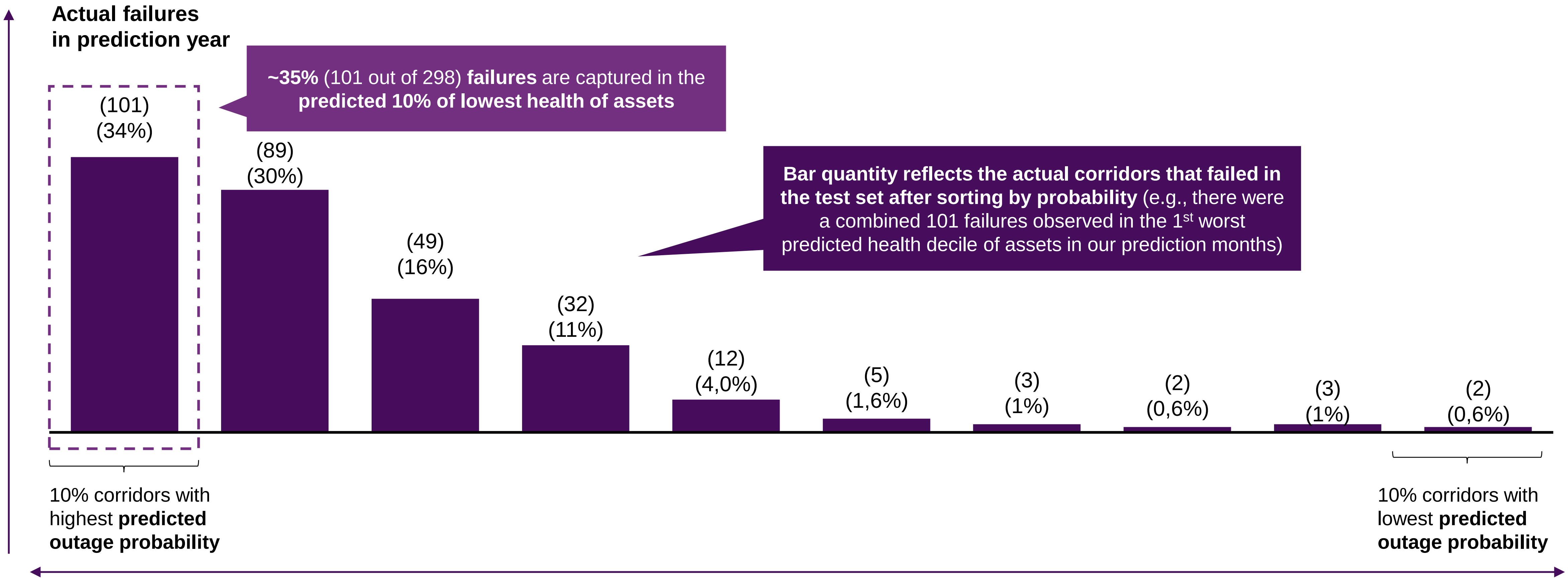
E. Decile Analysis on 2024 outages

Distribution of failures vs predictions for corridors in prediction set: 01/2024 – 10/2024



E. Decile Analysis on 2024 outages

Distribution of failures vs predictions for corridors in prediction set: 01/2024 – 10/2024



F. 60 features across 5 data sources contributed most to our preliminary model's predictive power

Corridor outage probability predictive contribution summary

Structural data | Outage data | Weather data | Trimming | Remote Sensing

Increases likelihood on increasing | Decreases likelihood on increasing

Contribution of data sources

Predictive contribution of source		Predictive features	
Structural	35%	5	Corridor length, types of lines
Outages	29%	6	Number of outages in the past years, type of outages
Weather	5%	28	Temperature, wind speed, precipitation
Trimming[1]	7%	16	Number of trimming activities in the past years
Remote Sensing	23%	6	Vegetation density
Total	100%	59	There are **60 features with predictive capacity over 71 features**

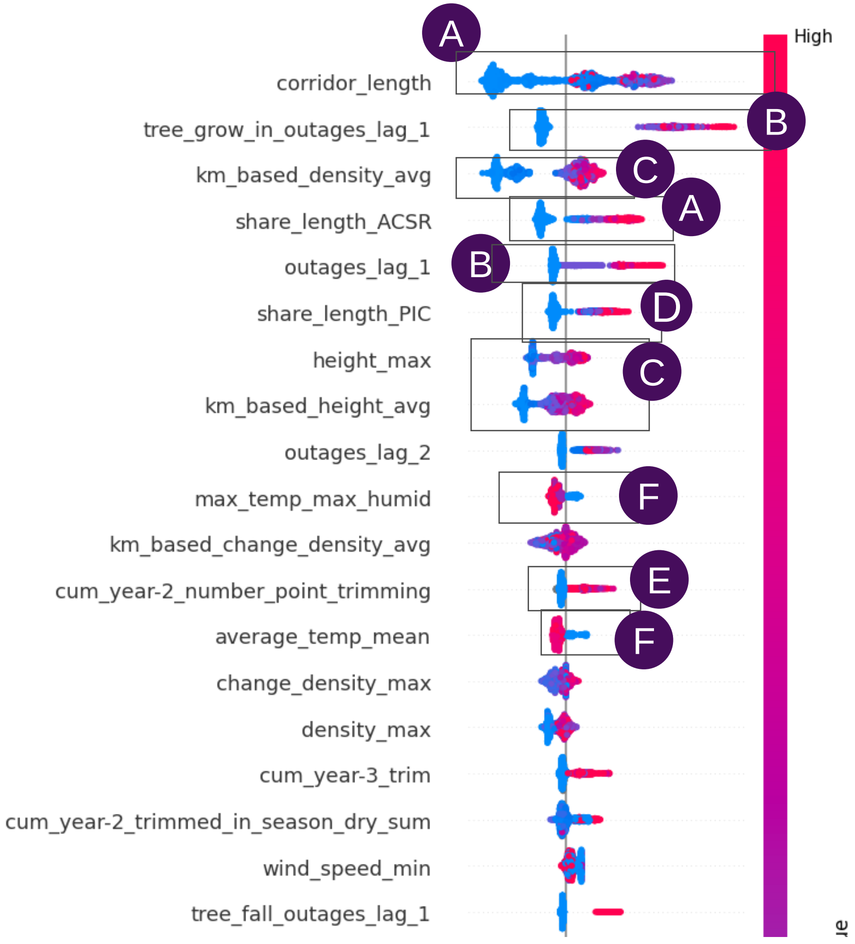
Feature importance, relative gain%

	Feature	Weight	Impact
1	Length of corridor	20.6%	
2	Share of Km of lines with aluminium conductor steel cable (ACSR)	18.6%	
3	Outages due to "tree grow in" in Y-1	8.9%	
4	Outages in Y-1	7.9%	
5	Average of vegetation density weighted by line length in Y[2]	6.5%	
6	Share of Km of lines with partial insulated cable (PIC)	5.3%	
7	Max of vegetation height in corridor in Y[2]	5.2%	
8	Average of vegetation density change weighted by line length in Y/Y-1[2]	3%	
9	Max of vegetation density in corridor in Y[2]	2.9%	
10	Average of vegetation height weighted by line length in Y[2]	2.9%	
11	Max of vegetation density change in corridor in Y[2]	2.8%	
12	Number of years with trimming in the past 3 years[4]	1.5%	
13	Outages in Y- 2	1.4%	
14	Maximum daily temperature in Y-1 in rain season [3]	1.4%	
15	Average daily temperature in Y-1 [3]	1.2%	
16	Outages due to "tree fall" in Y-1	1.2%	
17	Cumulative number of points trimmed in the past 2 years	1%	
18	Cumulative number of points trimmed in dry season in the past 2 years [4]	0.9%	
19	Cumulative number of points trimmed in the past 3 years	0.8%	
	Other 41 features	5%	

1. Currently including 66% of trimming data
2. Calculated as %pixel with vegetation in 10m buffer from line
3. Recorded in the closest station
4. Defined as unique devices reported in SAP work orders

F. We can see the directionalities of the features by aggregating the contributions for all corridors

Summary of model impact by feature value, SHAP[1]s



Increases likelihood of outage   Decreases likelihood of outage

Dimension	Impact on CM
A Long corridor with long length share of aluminium conductor steel cables	
B High number of outages in past years, most importantly outages due to tree growth	
C High km-weighted average tree density and maximum tree height	
D High share of corridor length made of partially insulate cables	
E Dryness of the area: high maximum and average temperature	
F Areas at high risk with multiple trimmings in the past years	

1. SHapley Additive exPlanations (name of the analyses)

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PEA

PROVINCIAL ELECTRICITY AUTHORITY

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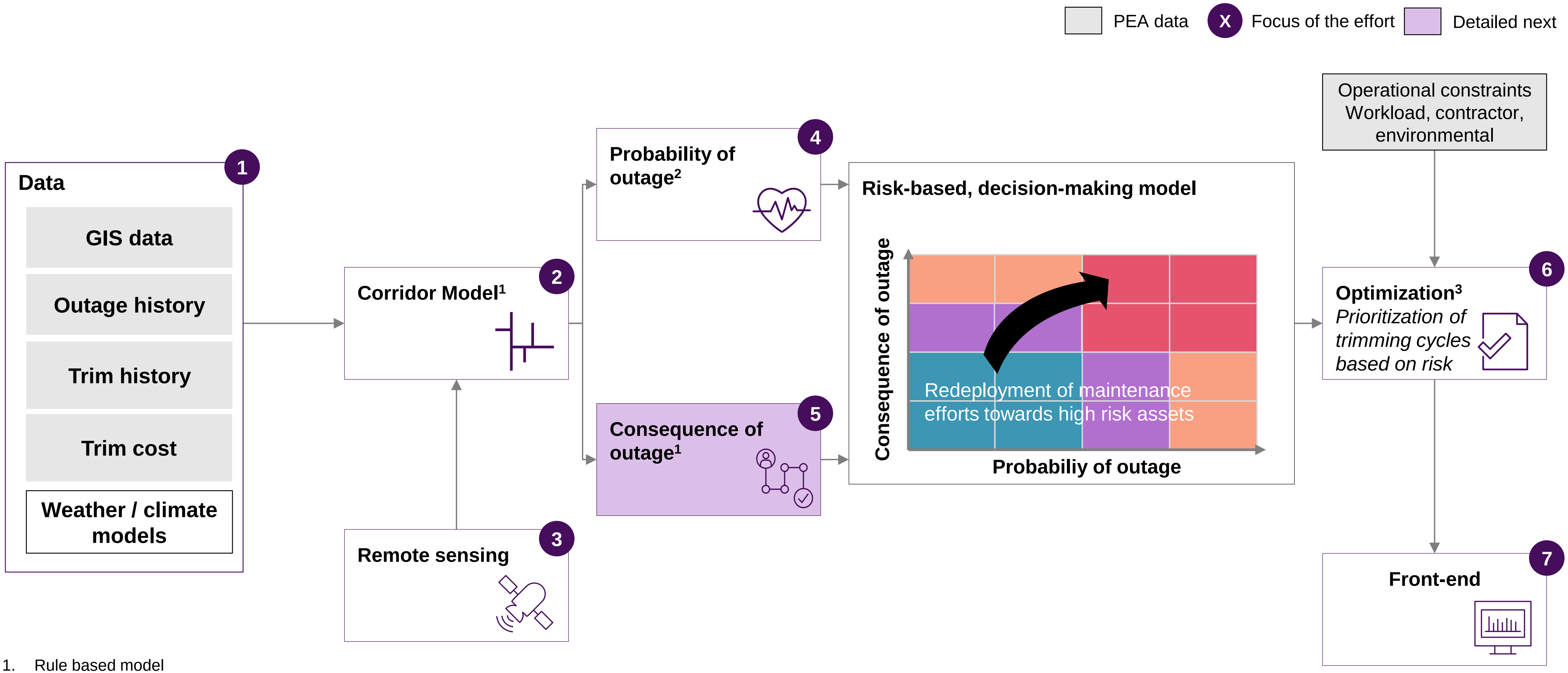


1. Rule based model
2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming

Source: McKinsey & Company

Downstream customers driven criticality is leveraged as the most relatable KPI to PEA operations

PEA data | Considered for future iterations | Selected option | X Detailed next

Option 1 Loss of profitability driven criticality

Calculation logic for criticality

Asset criticality index (THB)

- Service
- Mean repair cost [1]
- Fine
- Environ-ment
- Safety

Weight by customer type (feeder level)
- VIP (300%)
- Industrial (150%)
- Municipality (100%)
- Town (100%)
- Subdistrict (100%)
- Rural (100%)

Transformer type (kVA)
- PEA transformer (kVA)
- Customer transformer (kVA)

Capacity (kW)
- **Actual Capacity** (kW): Σ XX
- **Rated Capacity** (kW): Σ YY × **Capacity factor** (%): 50%

Revenue (BAHT/KWh)
- Tarif (BAHT/KWh): 3.9
- Tarif (BAHT/KWh): 4.2

Average outage duration (h)
- **Recloser 1**
- **Fuse 1.1**
- **Switch 1.2**

Option 2 Customer impact driven criticality

Asset criticality index (Customer interruptions)

Customers downstream

Imputation of null values proportional to load

Adjusted customers downstream

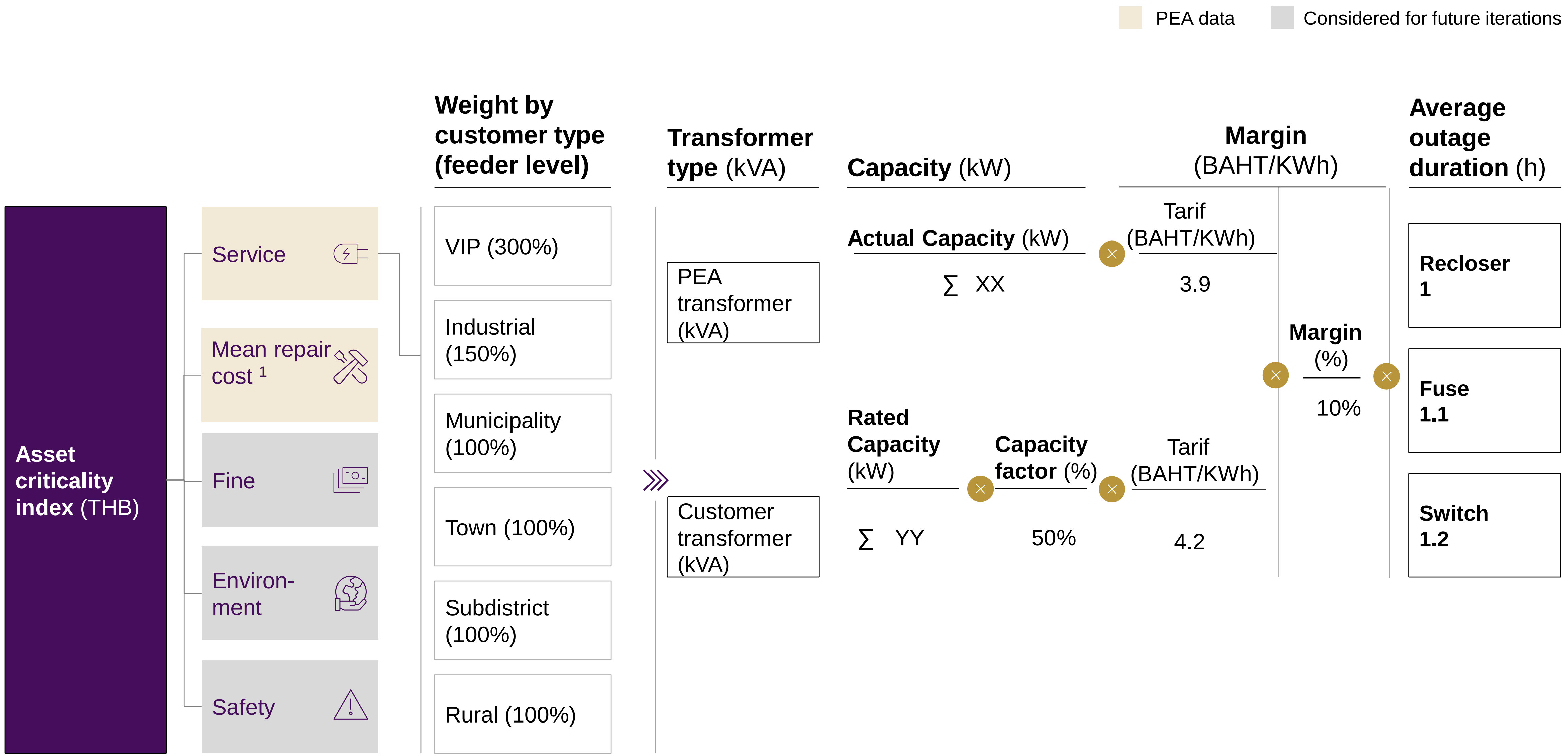
Risk metric

Option 1: **Risk is defined as loss of profitability in THB**

Option 2: **Risk is defined as customer interruptions** (expected outages* customers downstream) being **SAIFI the metric being optimized**

1. Calculated as corridor length * mean repair cost per meter, derived from PEA repair work orders

1A. Loss of profitability - cost as margin



1. Calculated as corridor lenght * mean repair cost per meter, derived from PEA repair work orders

1B. Loss of profitability - cost as revenue

PEA data | Considered for future iterations

Asset criticality index (THB)

- Service
- Mean repair cost [1]
- Fine
- Environ-ment
- Safety

Weight by customer type (feeder level)

- VIP (300%)
- Industrial (150%)
- Municipality (100%)
- Town (100%)
- Subdistrict (100%)
- Rural (100%)

Transformer type (kVA)

- PEA transformer (kVA)
- Customer transformer (kVA)

Capacity (kW)

- **Actual Capacity** (kW): ∑ XX
- **Rated Capacity** (kW): ∑ YY × **Capacity factor** (%): 50%

Revenue (BAHT/KWh)

- Tarif (BAHT/KWh): 3.9
- Tarif (BAHT/KWh): 4.2

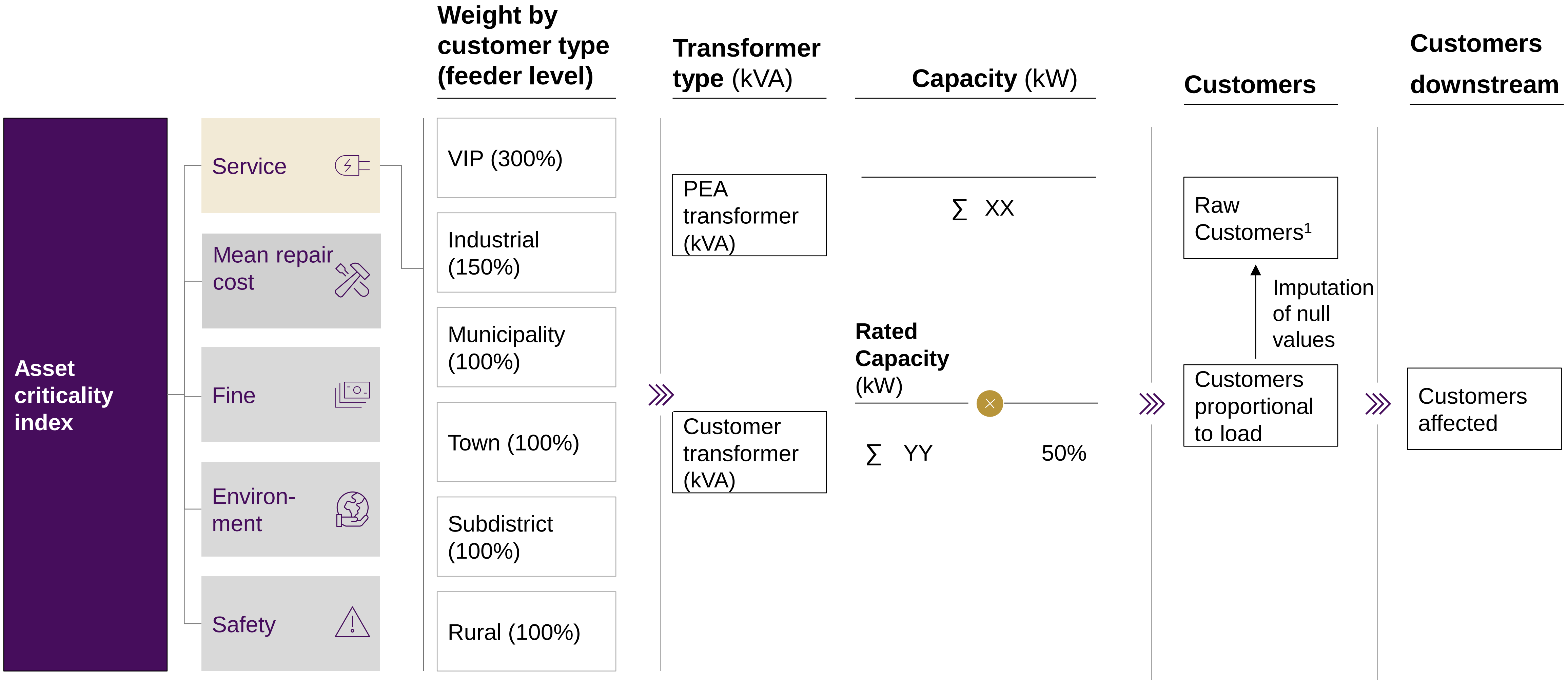
Average outage duration (h)

- **Recloser 1**
- **Fuse 1.1**
- **Switch 1.2**

1. Calculated as corridor lenght * mean repair cost per meter, derived from PEA repair work orders

2. (Chosen option) Customers downstream – driven

PEA data | Considered for future iterations



1. Derived from transformer layer

Criticality Concentration – options comparison

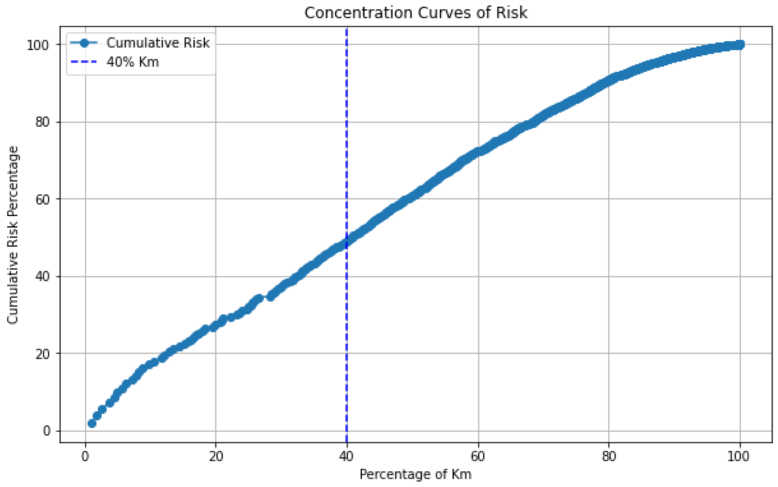
Option 1

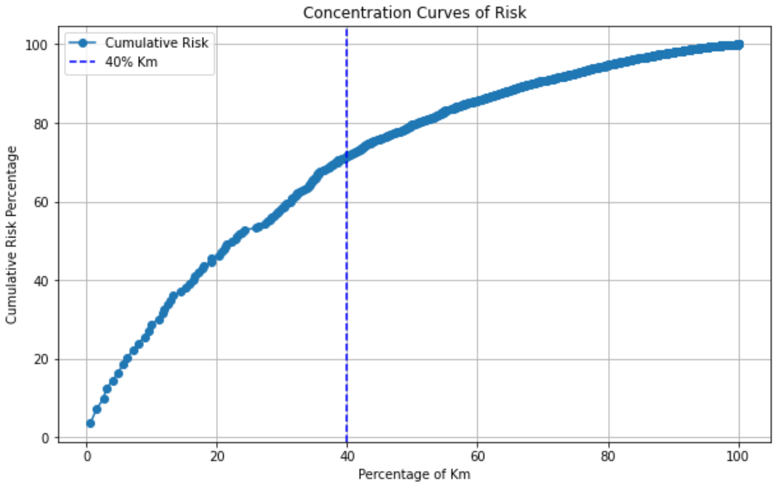
Option 2

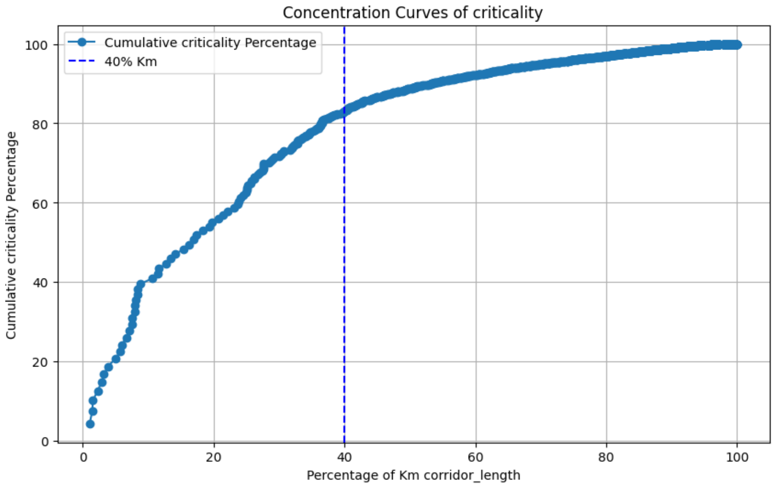
Criticality = Margin Lost + Repair

Criticality = Revenue Lost + Repair

Criticality = Downstream Customers







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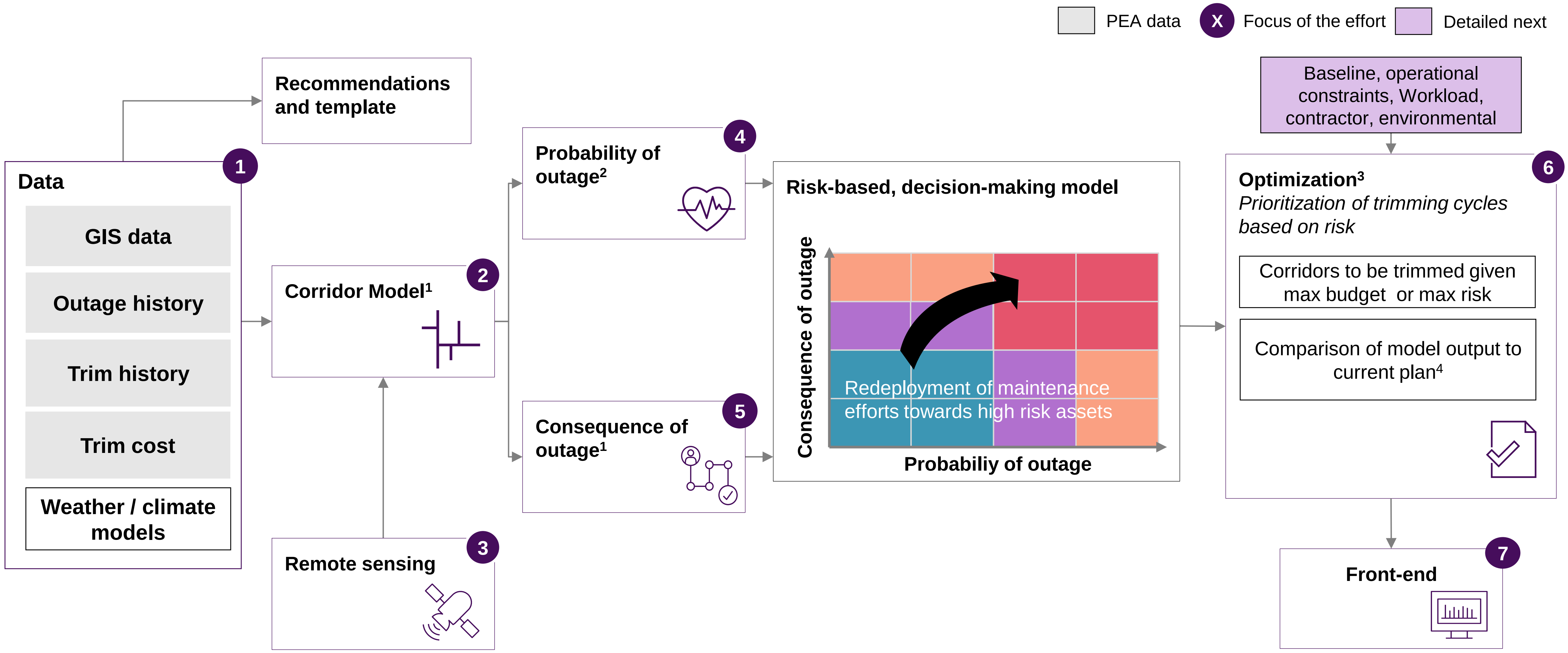
Our Vegetation Management approach leverages Advanced Analytics and remote sensing to optimize trimming cycles



1. Rule based model; 2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming; 4. Comparison consists of: trimming feeders, total cost, density of trees and therefore trimming cost by feeder

Source: McKinsey & Company

Trimming baseline activities follow a 3-year budgeting and operations cycle, which is not yet fully reflected in the data

Budgeting | **Operations**

Process

Y-2: Budget

The district team submit **mostly previous years' OPEX spend** to ask for a total budget for that district

The **central team reviews and allocated the budget**

Y-1: Trimming Plan

MJM

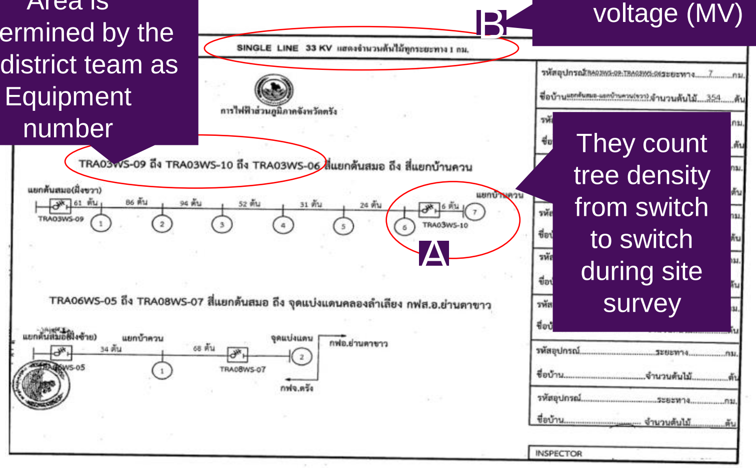
Sub-district teams receive budget and **rides through the line from protective device to protective device in their whole network and defines the trimming plan**

Once decided trimming is required, number of trees trimmed are counted per km and per protective device using MJM-linked phones with geo-tracking (MJMMFO)

Final budget is determined considering type of line and number of trees per km

Area is determined by the subdistrict team as Equipment number

33KV = Medium voltage (MV)



They count tree density from switch to switch during site survey

Y: Actual trimming

MJM

Prior to launching SAP Workorder (1 month to few days), **do another MJM survey in the same line to update the density and recalculate the middle price**

Launch PO on SAP using that middle price, then confirm vendors (usually the same vendors)

Trimming is carried out in the line sections defined on the plan

Trimming plan recorded in 2022/3 is available and corresponds to trimming activities carried in 2023/4

Lack of connectivity between SAP and MJM will be solved by end of year

Data

Trimming plan recorded in MJM (with price, density, and place).

Actual trimming is recorded in MJM

SAP records for financial tracking

Baseline

Comparison of trimming baseline cost

Method	Data Source	Level	Area: Feeders with PPA,PPB,TRA				Estimation
			Km coverage	% Coverage	Cost	Total	
1	SAP	Feeder	-	-	19.971.044		Raw Cost filtered by feeder
						19.971.044	
2	MJM	MJM line	2245	76,52%	11.954.437		Frequency: 2/year
	-	-	689	23,48%	3.668.867		Proportional to cost in km covered by MJM
						15.623.304	
3	META	GIS line	2934	100,00%	15.721.203		Density: META derived, Frequency: 2/year
						15.721.203	

Km Feeders
2.934

- **SAP:** total baseline of **20 M**
- **MJM:** 76% of network is trimmed. Assuming frequency of 2x/year and that the remaining 24% follows the same cost distribution, we obtain **~16M**
- **META:** Assuming frequency of 2x/year, **~16M**

Baseline Assumptions:

1) **Budget is 16 M**
2) **100% of the network** is trimmed
3) Frequency of trimming is **2 times a year**
4) **Density given by META** as it provides for the whole network

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While trimming model and operations show differences in granularity, a "level" for frequency optimization needs to be aligned to be operationally efficient for PEA: MJM WO, MJM line, GIS line or GIS corridor

Selected option X Detailed next

Category	Level
Outages	Protective Device
Health Model	GIS corridor
Criticality Model	GIS corridor
Optimized frequencies	GIS line
Trimming plan	MJM work order MJM line (for cost estimate)
Trimming execution	MJM work order

Connected through corridor model

Connected through corridor-line conversion model

Connected through MJM data recording

!

Difficulties in connecting them due to data differences

Recommendation

We can give the optimized frequency to a grouped recommendation at one of these levels:

1. **MJM Work Order**
2. **MJM Line**
3. **GIS line**
4. **GIS Corridor**

Consider 2 steps between:

1. **Trimming interval instruction**
2. **Bundling of instructions for contracting**

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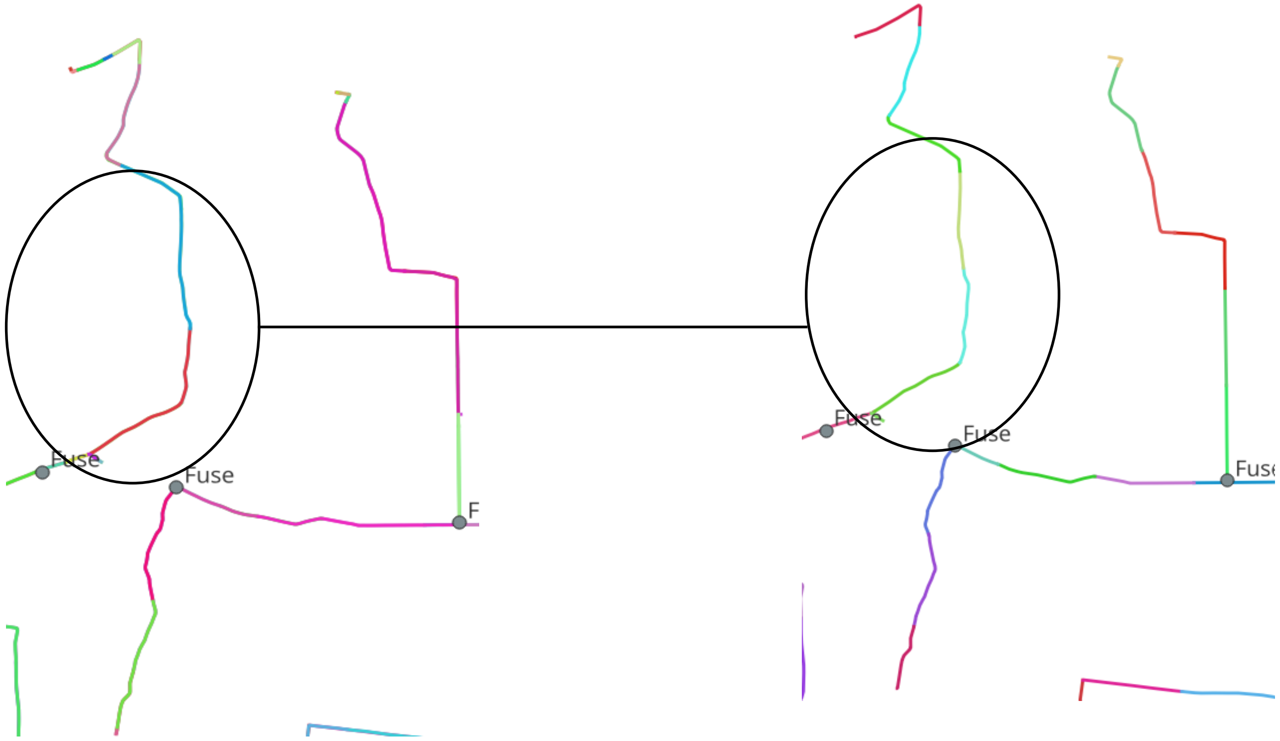
Connecting MJM and GIS data is difficult due to different definitions and data recording issues

MJM definition issues

MJM lines are not defined as between protective devices

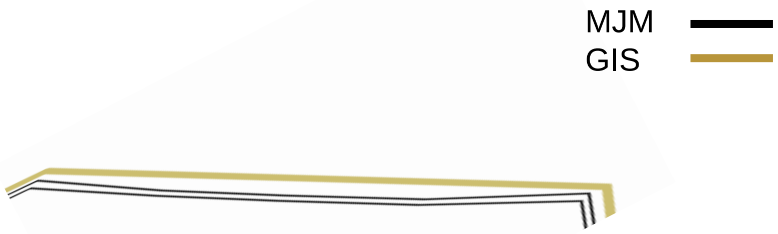
MJM and GIS lines have different granularity

MJM **GIS**

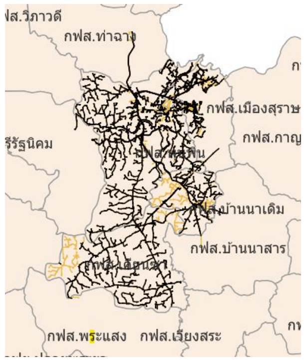


MJM data issues

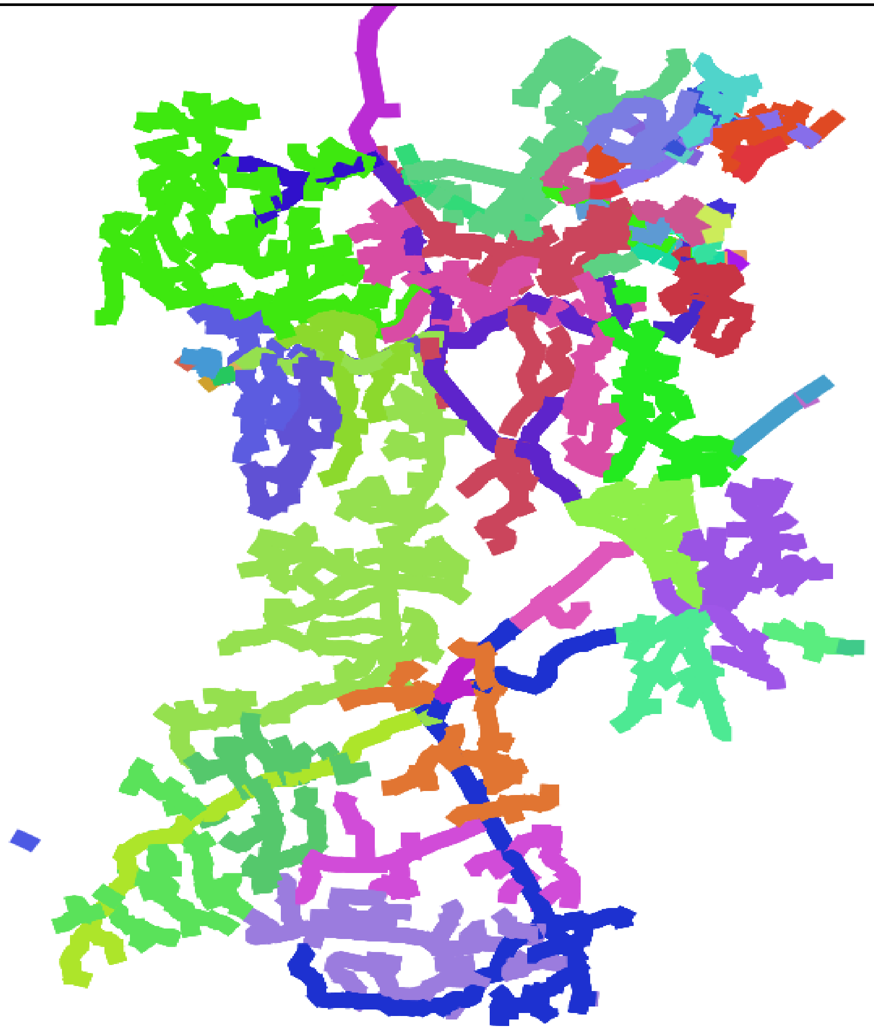
MJM lines have an offset and do not overlap and intersect GIS lines



There are missing MJM lines (30% of scope)



❶ Granularity: MJM Work Order



Metrics

Average of length: 6.8 Km

Standard Deviation of length: 17.3 Km

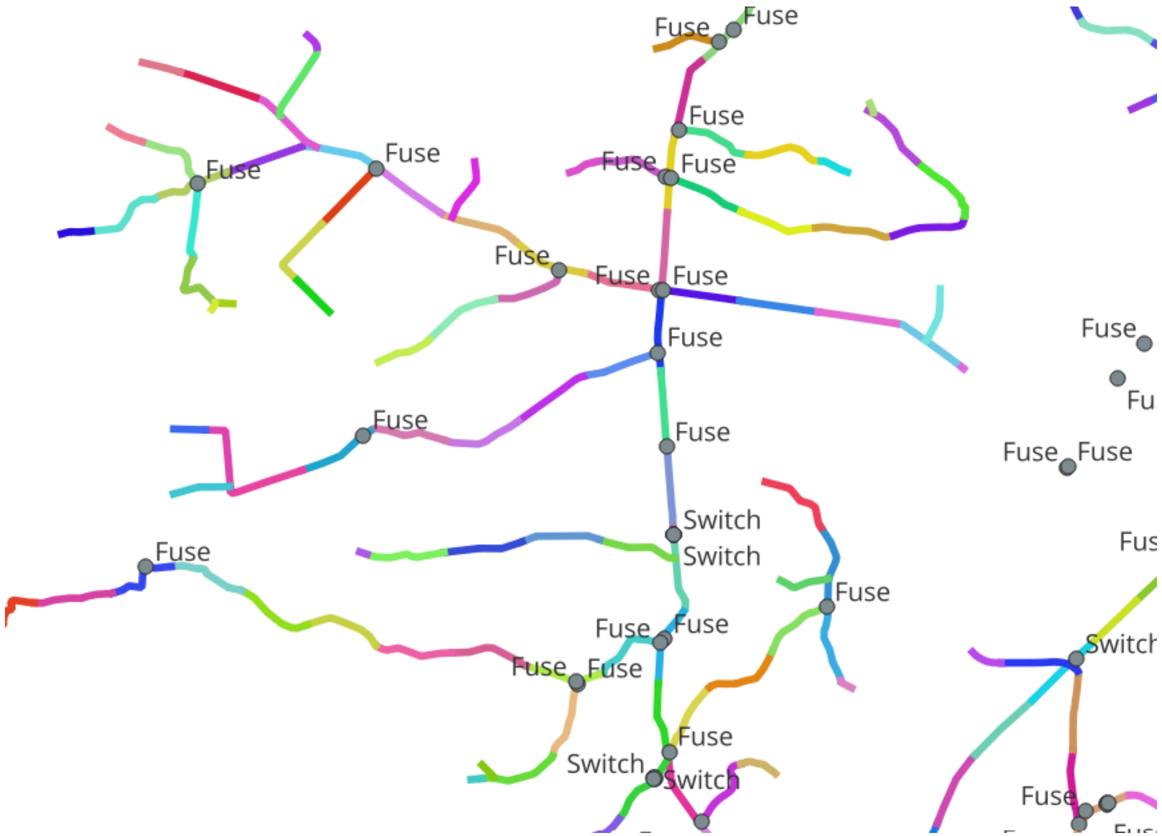
Advantages

- Operationally consistent with current practices

Disadvantages

- Very low granularity
- Model outputs (health, criticality) are at GIS corridor or line level – most of the value deriving from the models is lost due to GIS-MJM mapping approximation and Work Order level grouping
- Only covers ~70% of the scope
- Unstable in time (changes from execution to execution)

❷Granularity : MJM Line



Metrics

Average of length: 0.75 Km

Standard Deviation of length: 0.33 Km

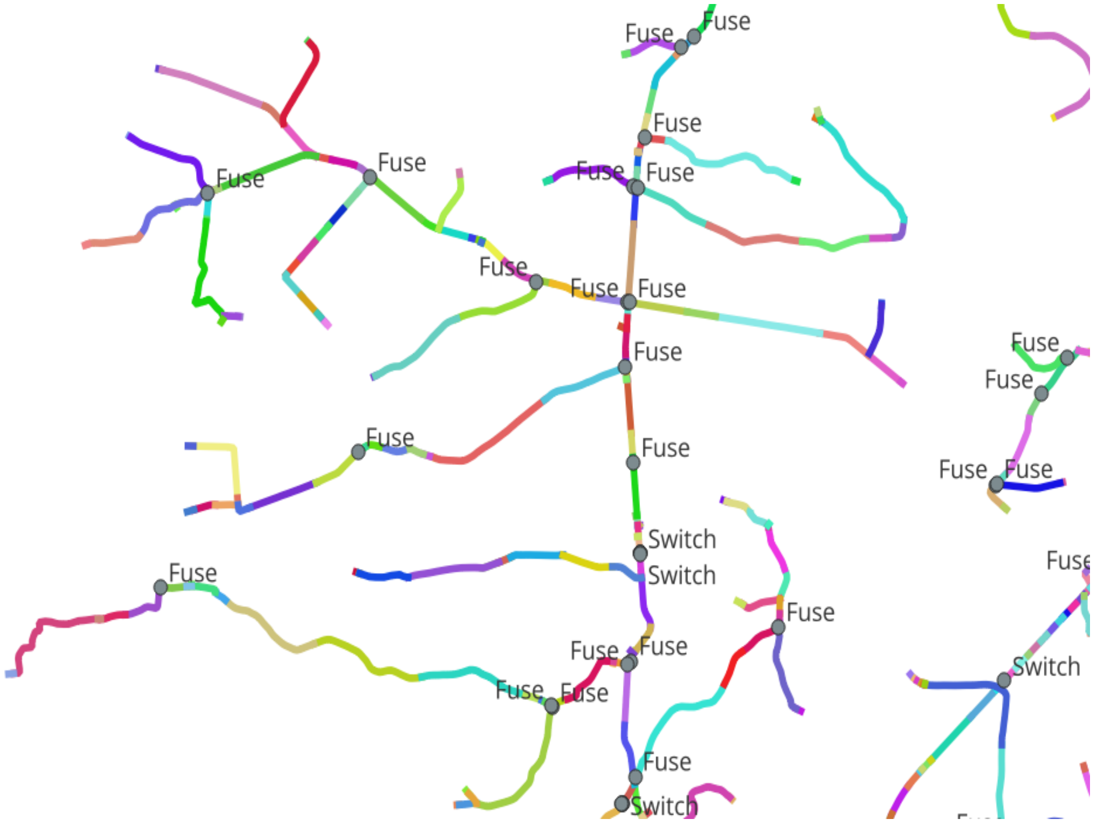
Advantages

- Operationally consistent with current practices
- High granularity

Disadvantages

- Model outputs (health, criticality) are at GIS corridor or line level – part of the value deriving from the models is lost due to GIS-MJM mapping approximation
- Only covers ~70% of the scope
- Unstable in time (changes from execution to execution)

❸ Granularity : GIS Line



Metrics

Average of length: 0.111 Km

Standard Deviation of length: 0.256 Km

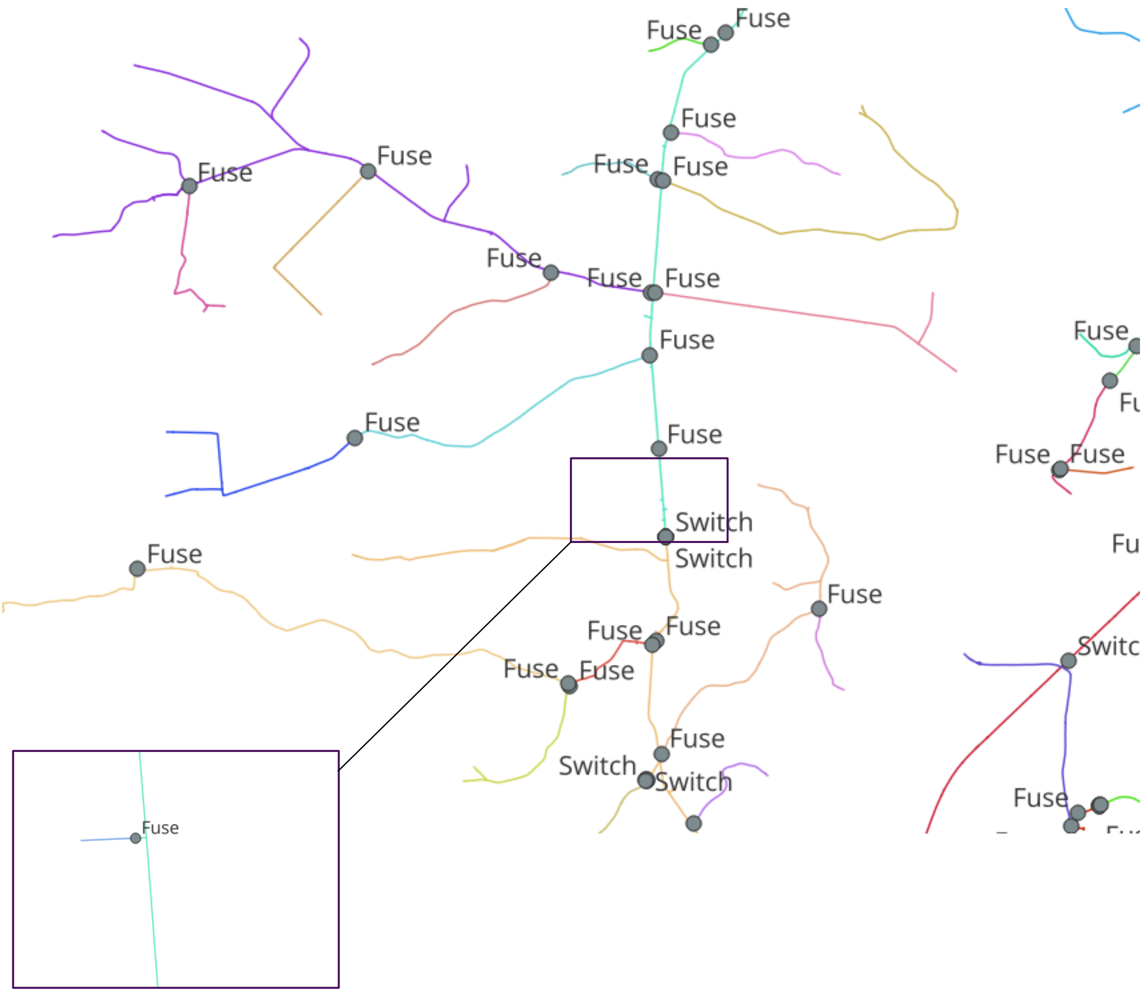
Advantages

- High granularity
- Captures full value of the model: model outputs (health, criticality) are at GIS corridor or line level

Disadvantages

- **Requires major change in operational practices**

❹ Granularity : GIS Corridor – selected option



Metrics

Average of length: 1.8 Km

Standard Deviation of length: 3.9 Km

Advantages

- Clear definition: between protective devices
- Captures full value of the model: model outputs (health, criticality) are at GIS corridor or line level
- High visibility: easy to monitor model results by connecting outages to recommendation

Disadvantages

- **Requires minor change in operational practices**

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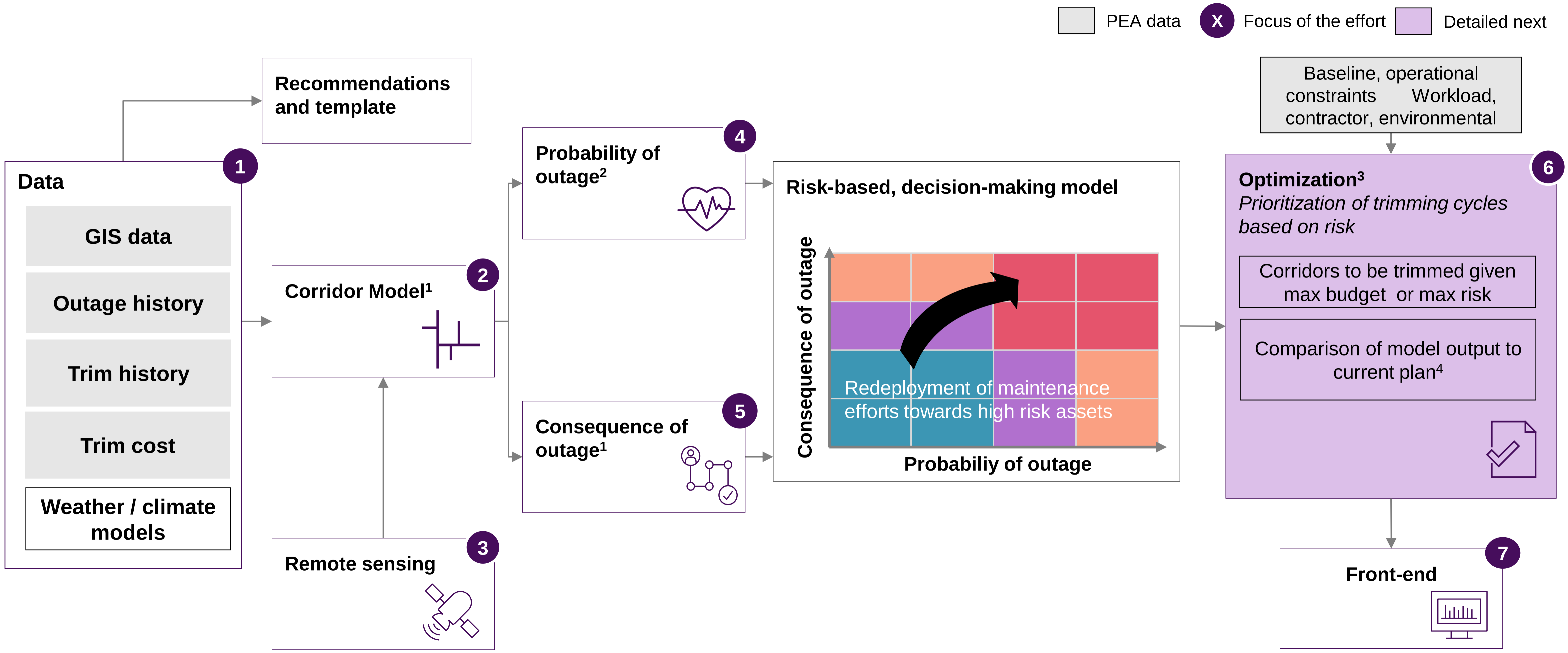
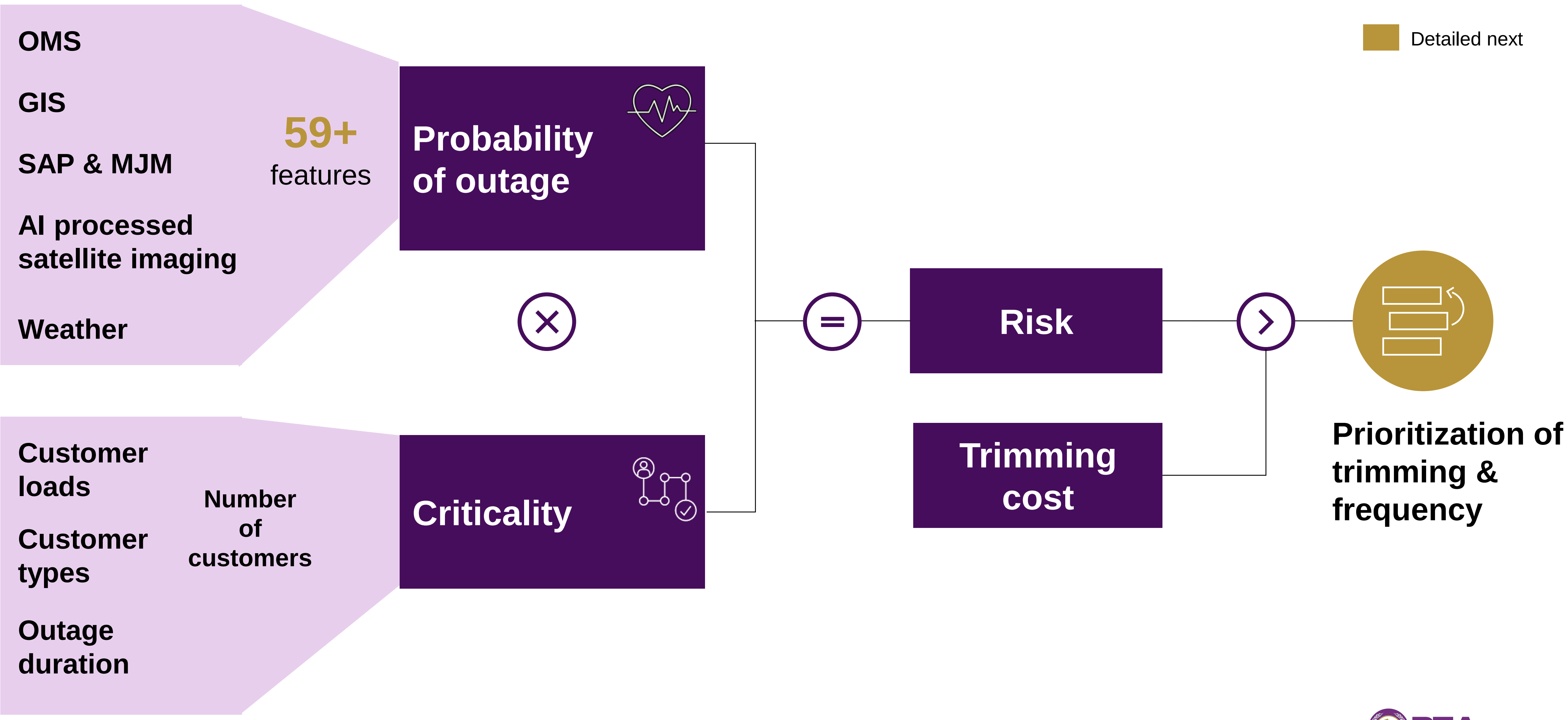
Our Vegetation Management approach leverages Advanced Analytics and remote sensing to optimize trimming cycles



1. Rule based model; 2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming; 4. Comparison consists of: trimming feeders, total cost, density of trees and therefore trimming cost by feeder

Source: McKinsey & Company

Simplified risk-based vegetation management concept



Trimming optimization consists of transitioning from same trimming-cycles for all corridors to a customized frequency for each based on risk and cost

Selected optimal frequency – methodology detailed next

	Feeder 1		
	Corridor 1	**Corridor 2**	**Corridor 3**
Density (RS approach)	Low	Medium	High
Cable Material (GIS Data)	Partially Insulated	Partially Insulated	Aluminum
Length (GIS Data)	1.8 km	2.7 km	3.5 km
Risk	$Risk_1$	$Risk_2$	$Risk_3$
Cost (based on density)	Cost (low density)	Cost (medium density)	Cost (high density)
Baseline trimming frequency	2x/year	2x/year	2x/year
Optimized trimming frequency			
Frequency 1	1x/year (selected)	1x/year	1x/year
Frequency 2	2x/year	2x/year (selected)	2x/year
Frequency 2	3x/year	3x/year	3x/year (selected)

Optimal frequency for each corridor is selected based on its investment return, given by the risk removed and cost

Procedure

1. For each corridor and frequency, investment return is determined as the ratio between the risk you are able to remove, and the cost implied

Investment return (IR) = **Risk reduction** / **Cost of trimming**

2. First, algorithm selects the lowest frequency for all corridors. In this case, 0.5x/year = every 2 years. The total required budget is determined

3. Then, corridor-frequencies are ranked by return. A corridor will be "changed" to the following frequency only if budget allows for it and the return compared to previous frequency is higher

$$\text{DiffI}_R = \frac{\text{Risk reduced}_T}{\text{Cost of trimming}_T} - \frac{\text{Risk reduced}_{T-1}}{\text{Cost of trimming}_{T-1}}$$

4. Algorithm stops either when maximum budget or expected risk reduction are achieved

Result: list of optimal frequency for each corridor

Example

Detailed next

Corridor	Frequency	Risk removed	Cost	Return = risk removed / cost
C1	1x/year	RR_2	cost	→ IR_2 (C1 – 1x/year)
C1	2x/year	RR_3	2*cost	→ IR_3
C1	3x/year	RR_4	3*cost	→ IR_4

Risk removed is the amount of risk that is reduced when trimming, and increases the more frequently you trim vegetation; RR_1, <…<RR_5

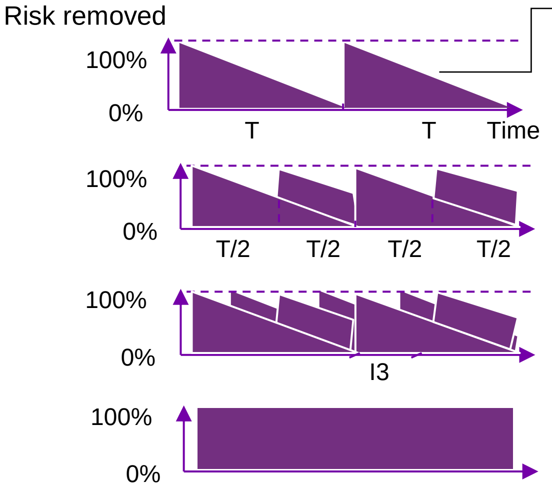
Risk removed by trimming at each frequency depends on the degradation time of the asset and inspection frequency

X Detailed next

Risk removed by trimming at each frequency

The **risk removed depends on the degradation time** of the asset **and inspection frequency**, as failures can happen between trimming if inspected less frequently

1. We determine how much risk can be removed from the corridor with trimming, assuming we go every day: **max risk**
2. **Degradation time (T)** is determined: **6 months** for vegetation related issues in corridors
3. **Risk removed at each frequency** is determined comparing the frequency to the degradation time

Frequency	Rationale	Frequency factor	Risk removed
T		50%	RR_T = 50%*max_risk
2T		75%	RR_2T = 75%*max_risk
3T		87%	RR_3T = 75%*max_risk
Every single day		100%	RT_ everyday = 100%*max_risk = max_risk

Covered area in purple represents how much of max risk can be removed at that frequency, or "**frequency factor**"

$Ff(n) = \sum_{i=1}^{n}(50\%)^n$,
n= frequency(T)

For any frequency, the risk removed is determined as: (max risk)* (frequency factor)

1. Leveraging an FMEA generated through GenAI, we determine how much risk can be removed from the corridor with trimming

Total Risk	**5,770**
Risk reducible by trimming	**2,234**
Max risk reduced	**40%**

Subsystem	Failure Mode	Effect	Severity (1-10)	Cause	Occurrence (1-10)	Prevention/Mitigation Actions Suggested	Detection (1-10)	Reducible by trimming	Reducable Risk	Reducible Risk by trimming	Total Risk
Line	Tree contact	Power outage due to line interruption	9	Vegetation growth	9	Trimming/Inspection of vegetation	8	1	648	648	810
			9	Branch/tree fall due to vandalism	3	Regular surveillance of surrounding areas	6	0	162	0	270
			9	Branch/tree fall due to extreme weather/wind	3	Installation of weather monitoring systems	1	0	27	0	270
			9	Branch/tree fall due to seasonal weather/wind	8	Trimming/Inspection of vegetation	7	1	504	504	720
			9	Branch/tree fall due to construction activities	5	Coordination with construction activities	4	0	180	0	450
			9	Tree fall due to landslides	4	Landslide risk assessment	3	0	108	0	360
			9	Tree fall due to disease	3	Monitoring of tree health	2	0	54	0	270
Protective Device (Switches, Fuse, Recloser)	Physical damage	Inability to protect circuit, causing outages	9	Vegetation growth	7	Trimming/Inspection of vegetation	8	1	504	504	630
			9	Branch/tree fall due to vandalism	1	Enhanced security measures	5	0	45	0	90
			9	Branch/tree fall due to severe weather/wind	3	Installation of protective barriers	1	0	27	0	270
			9	Branch/tree fall due to seasonal weather/wind	6	Trimming/Inspection of vegetation	7	1	378	378	540
			9	Branch/tree fall due to construction activities	3	Coordination with construction activities	4	0	108	0	270
			9	Tree fall due to landslides	2	Landslide risk assessment	3	0	54	0	180
			9	Tree fall due to disease	1	Monitoring of tree health	2	0	18	0	90
Crossarm / Poles	Inadequate vegetation clearance	Contact with vegetation, causing outages	5	Vegetation growth	5	Trimming/Inspection of vegetation	8	1	200	200	250
			5	Branch/tree fall due to vandalism	1	Enhanced security measures	5	0	25	0	50
			5	Branch/tree fall due to severe weather/wind	2	Installation of weather monitoring systems	4	0	40	0	100
			5	Branch/tree fall due to construction activities	1	Coordination with construction activities	4	0	20	0	50
			5	Tree fall due to landslides	1	Landslide risk assessment	3	0	15	0	50
			5	Tree fall due to disease	1	Monitoring of tree health	2	0	10	0	50

Source: Leveraged model is GPT 3.5

3. Risk removed at each frequency is determined comparing the frequency to the degradation time and using max risk value

Frequency	Degradation time (T)	Frecuency (Times T)	Max risk	Frequency factor	Risk removed
1x / year		0.5T		29%	11%
2x / year	6 months	T	40%	50%	20%
3x/ year		1.5T		65%	26%
Every single day				100%	39%

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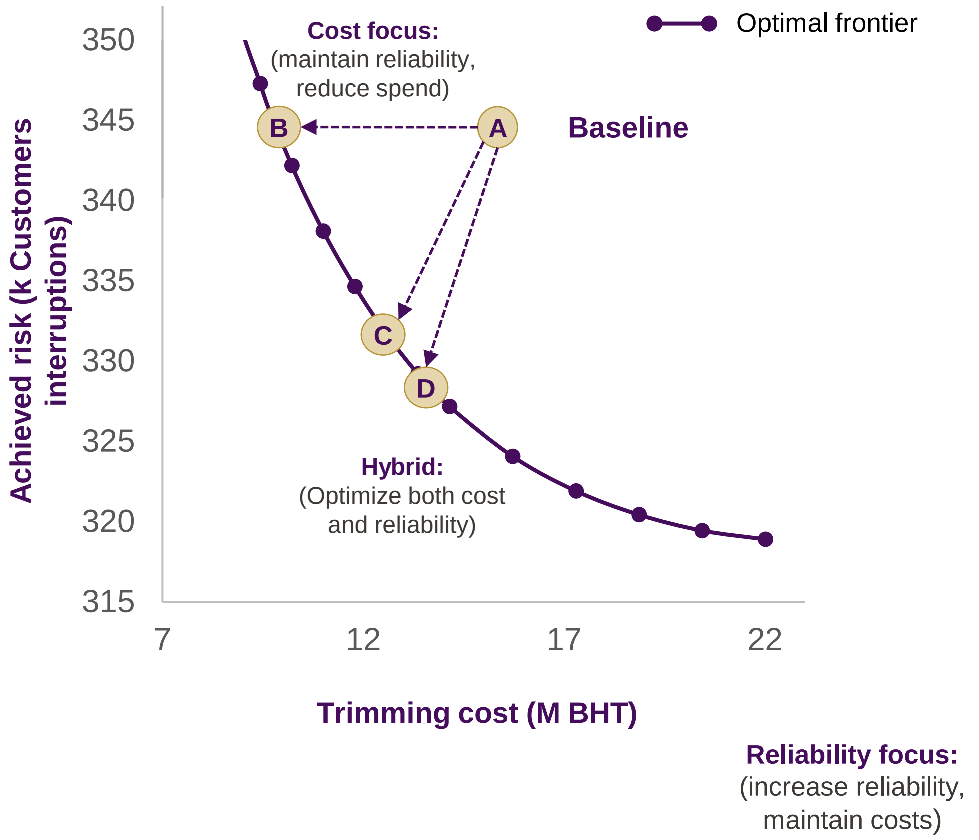
Trimming prioritization: SAIFI and cost are optimized by moving from fixed time-based trimming cycles to risk-based frequencies for each corridor

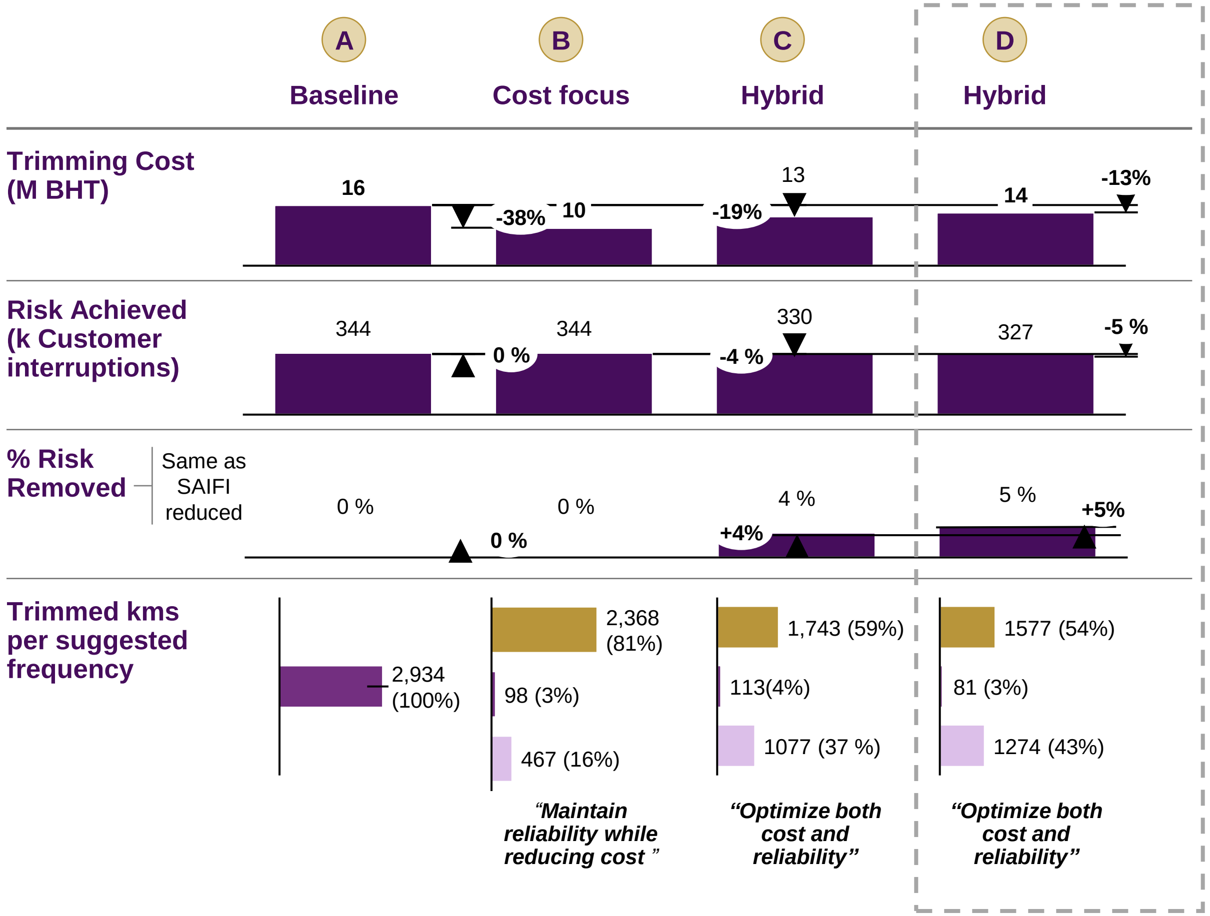
PRELIMINARY

Detailed next

Trimming Cost – Achieved Risk curve



Comparison



Hybrid model suggests increasing frequencies for high-risk corridors and decreasing it for low-risk, in order to improve SAIFI and reduce costs

ILLUSTRATIVE

A Baseline: time-based trimming 2-time per year

	Length, km	Corridor risk	Avg tree density (in-person survey)	Trimming interval
Corridor 1	7	Unknown	High	2x/year
Corridor 2	8	Unknown	High	
Corridor 3	9	Unknown	Medium	
Corridor 4	10	Unknown	Low	
Corridor 5	12	Unknown	Medium	
Corridor 6	10	Unknown	High	
Corridor 7	5	Unknown	Low	

D Hybrid model: risk-based trimming

Corridor risk	Avg tree density		Trimming interval	
High	High		3x/year	Increased interval based on risk
Medium	Medium	Validated tree density from HQ	2x/ year	
Medium	Low		2x/ year	
Low	Low		1x/ year	Extended interval based on risk
Medium	Low	Validated tree density from HQ	1x/ year	
High	High		3x/ year	Increased interval based on risk
Low	Low		1x/ year	Extended interval based on risk

	Baseline		Hybrid model		Saving, mnTHB
Phunphin & Trang (2,934 km)	15,721,203	−	13,693,167	=	2 mn
All MV lines (301K km)	1,031,458,065	−	897,368,516	=	134 mn

+5% Improvement in SAIFI

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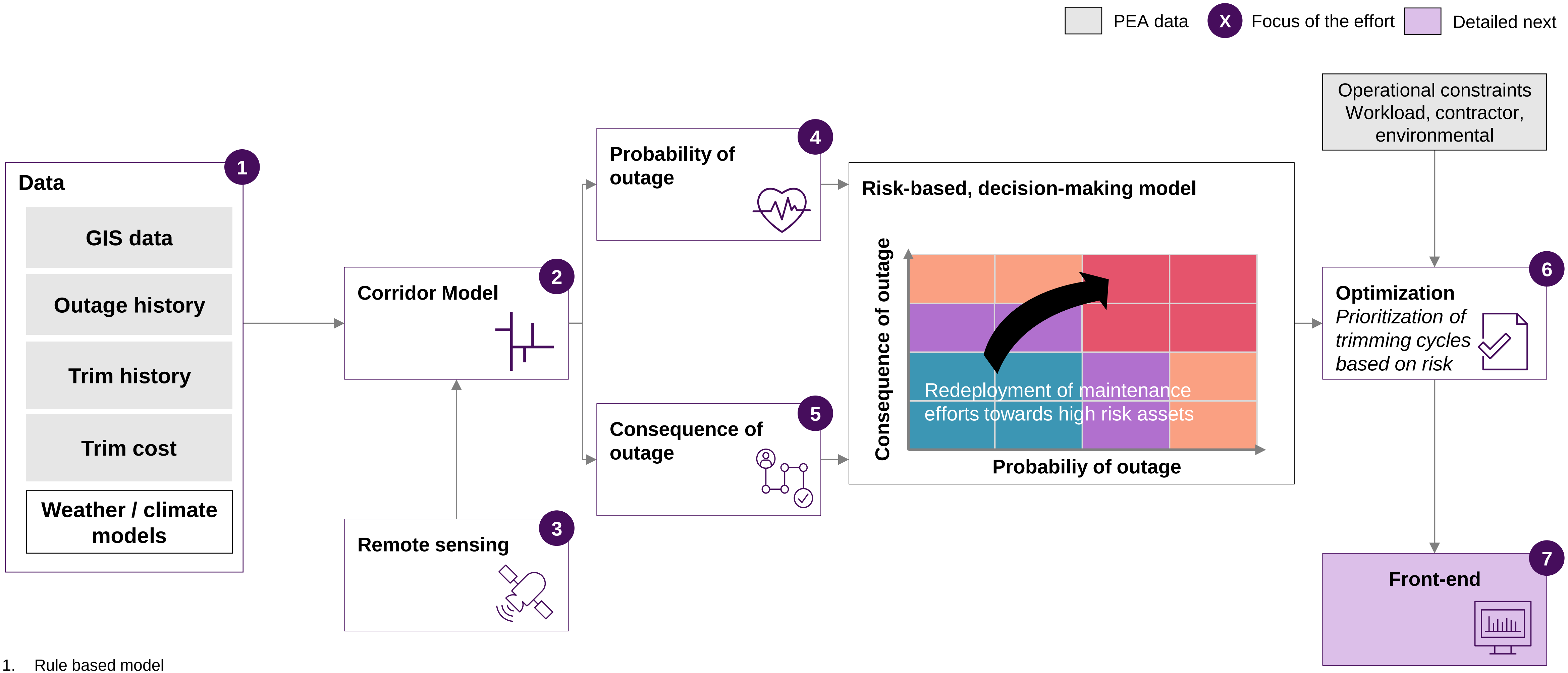


1. Rule based model
2. Advanced Analytics model - supervised Machine Learning model for classification: e.g., random forest, lightGBM
3. Mixed Integer Programming

Source: McKinsey & Company

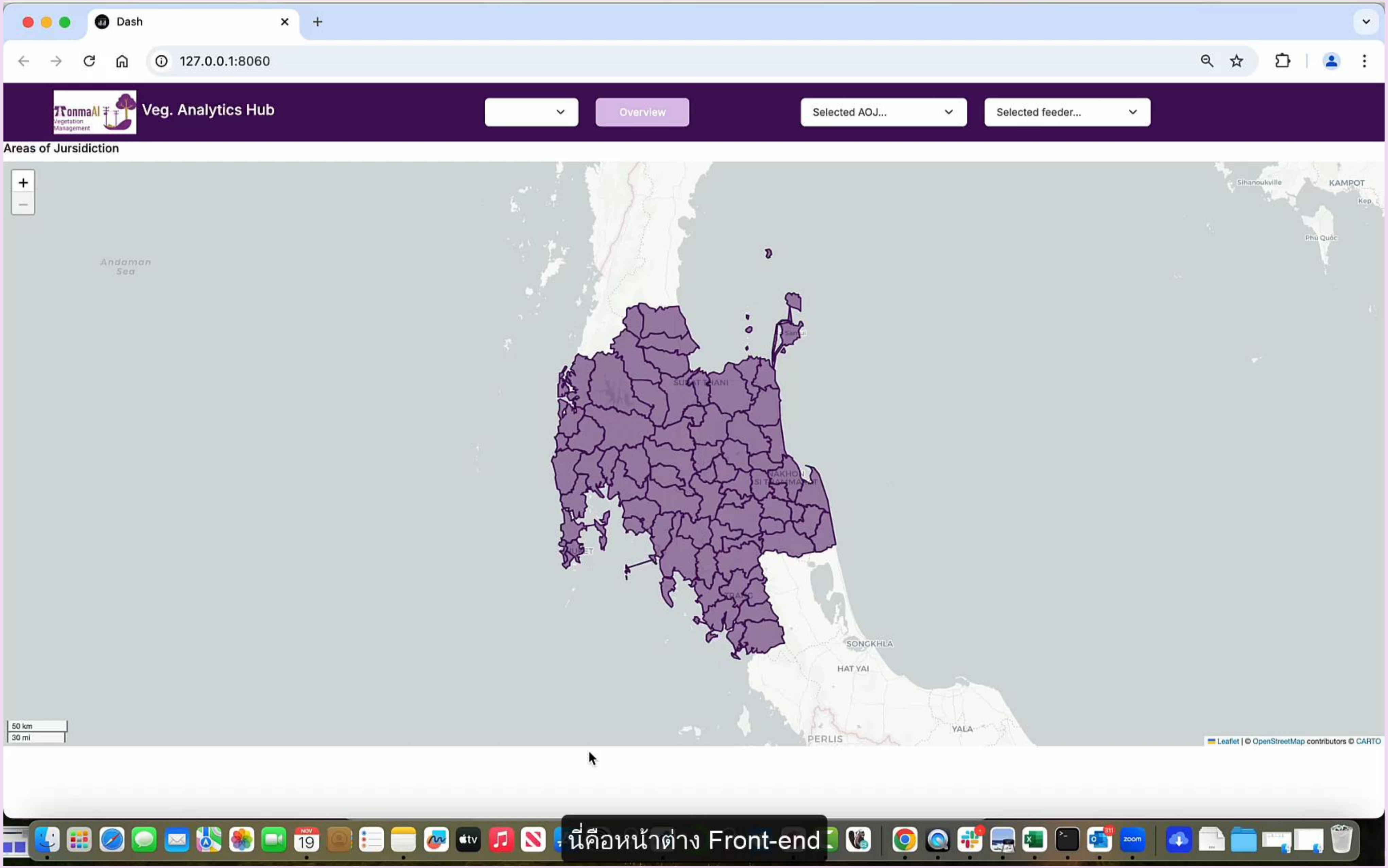
We have analyzed the characteristics and advantages/disadvantages of different visualization tools, concluding with ArcGIS as the preferred option

Aligned option

	PowerBI 1	ArcMap 2	ArcGIS 3	Web app 4
Description	Create **visualization dashboard** with health, criticality and risk analysis **and geospatial layers with ArcGIS integration**	Build **geospatial layers** on collected data, with health, criticality and risk maps and **include them on PEA's ArcMap system**	Build **visualization dashboard** with health, criticality and risk analysis and **interactive map**	**Develop web-app and implement it in PEA's system.** Dash or other externally developed web framework
Map	**Only one layer supported** without ArcGIS licence Small size	**Supports multiple layers types** and superposition	**Supports multiple layers** types and superposition **High quality**	**Supports multiple layers** types and superposition **High quality**
Dashboard	**Highly customizable graphs**, data interconnection and interactive elements	**No dashboard**	**Highly customizable** graphs, data interconnection and interactive elements	Can include a **limited set of predefined charts**
Capabilities required	No extra resources required	No extra resources required	No extra resources required	**Requires data science and web development capabilities** to update and maintain
Cost	**License** PEA's ArcGIS license, sharable?	**License** PEA's ArcMap license, sharable?	**License** PEA's ArcGIS license, sharable?	**License** Open Source
Advantages	PEA's **familiarity with PowerBI** Easy modifications and interactions Connectivity between map and tabular data sources	**Directly integrated in PEA's ArcMap system** **Highly flexibility** for in-tool analysis	**Directly integrated** in PEA's ArcGIS system **Easy modifications and interactions** Connectivity between map and tabular data sources	**Highly flexible and customizable** Tuned to the vegetation management use case
Dis-advantages	Limited quality and size of the map Performance may slow down for large geospatial data	Limited user experience No interaction between geospatial layers and other data / analysis	Performance may slow down for large geospatial data	Possible challenges in integration and compatibility with PEA's system

E. Front-end

Web-based Dash



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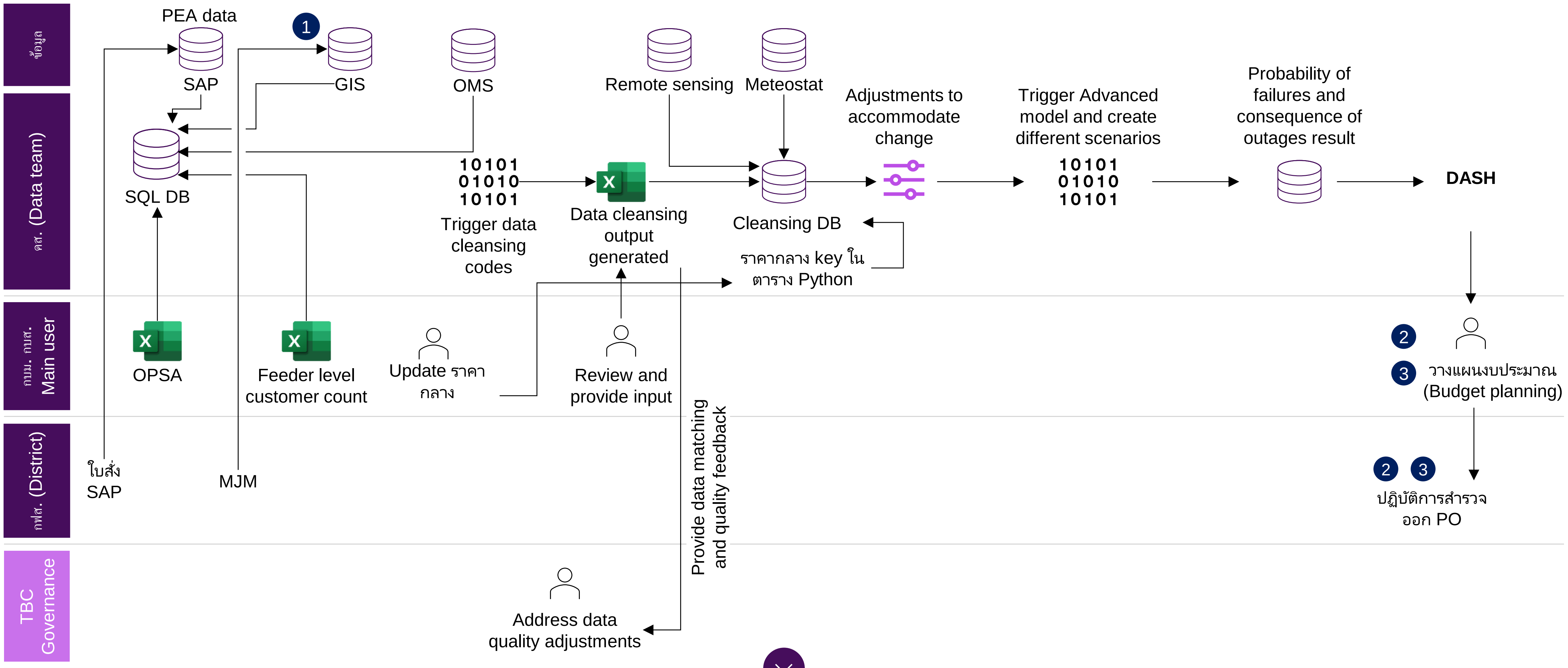
Process change: Vegetation Management tool will improve vegetation planning, enhance collaboration and visualization for better decision making



Process Change

1 **MJM/GIS update**

2 **Procurement:** กระบวนการจัดซื้อจัดจ้าง

3 **Trimming practice:** ระเบียบการวางแผนตัดต้นไม้

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Process Change Plan Updates

	Item	Process owner	Progress	Next steps
1 **MJM/GIS**	**1. Update MJM** – MJM สามารถระบุ หมายเลขใบงาน SAP เพื่อการอ้างอิงที่ถูกต้อง – การสำรวจ MJM มี field ที่ระบุชนิดของพืชพรรณต้นไม้ – การลง MJM เพื่อจัดทำแผน และจัดซื้อให้ วาดเส้นทางสำรวจโดยอิงจาก line ที่อยู่ภายใน corridor แทนที่การวาดเอง **2. Update GIS Line** – **Offset GIS line** ให้ตรงกับแนวสายไฟจริง	ESRI/GIS กบร.	• ปรึกษาหารือกับ Esri และ GISเมื่อวันที่ 12 พย. และอยู่ระหว่างจัดเตรียมข้อสรุป	• GIS: จัดทำเอกสารสรุปข้อสังเกตุให้พิจารณาในแผนในวาระอันสมควร • MJM: จัดทำเอกสารสรุปเรื่อง from-to และ แนวทางการปรับปรุงระเบียบ (หากมี) • นำเสนอเขต • นำเสนอกบร. รับทราบ และแก้ระเบียบ
2 **Procurement**	• สำหรับเส้นทางที่มีการตัดแต่ง 3 ครั้งต่อปี (จากเดิมสองครั้งต่อปี) ให้ดำเนินการจัดซื้อแบบสัญญารายปีกับผู้รับเหมา	กบร.	• มีแผนการตัดตามโมเดลที่แนะนำ ควรตัดประมาณกี่ครั้งต่อปี โดยบางแนวทางแนะนำให้ตัดมากที่สุดไม่เกินสามครั้งต่อปี	• ศึกษาระเบียบการวางแผนตัดต้นไม้ • จัดทำเอกสารสรุปเรื่อง from-to และ แนวทางการปรับปรุงระเบียบ (หากมี) • นำเสนอเขต • นำเสนอกบร. รับทราบ และแก้ระเบียบ
3 **Trimming practice**	• การดำเนินงานและวางแผนอิง risk-based plan ที่ตัดตรงพื้นที่ที่มีความเสี่ยงสูงก่อน • วางแผนช่วงเวลาการตัดให้พื้นที่ที่อยู่ใน corridor เดียวกันถูกตัดพร้อมๆกัน	กบร.		• ศึกษาระเบียบการวางแผนตัดต้นไม้ • จัดทำเอกสารสรุปเรื่อง from-to และ แนวทางการปรับปรุงระเบียบ (หากมี) • นำเสนอเขต • นำเสนอกบร. รับทราบ และแก้ระเบียบ

PEA PROVINCIAL ELECTRICITY AUTHORITY

Vegetation: 5 years roadmap

Preliminary

	2025	2026	2027	2028	2029
	Scale-up *Scale process changes and digital tools to all regions*		**Enhanced** *Update the model with additional data sets (collected during scale-up) and new remote sensing techniques including drones*		**Sustain** *Embed fully to business as usual*

Continuously test if current interventions are working, collect feedback on how to improve, and implement changes

Data recording
- 100% MJM recording with validation against SAP
- Standardize grouping of data across regions, e.g., set of common species
- Refresh model with complete set of data points collected
- Mapping of GIS line and MJM line for accurate mapping of data

Model update & refresh
- Expand inputs to cover Central & South
- Collect feedback & refine features
- Expand inputs to cover North & Northeast
- Collect feedback & refine features
- Annual model validation
- Annual model validation
- Annual model validation

Remote sensing
- Utilize readily available sensing data, e.g., Meta & Sentinel
- Incorporating new technologies, e.g., remote sensing & drone technology
- Explore alternative RS, e.g., other sat imaging providers & drones
- **Expand use-case to validate MJM tree counts and cost estimate**

Trimming practices

PPA, PPB & TRA
- Ensure compliance to risk-based plan
- Monitor result
- Risk elimination trimming: address whole corridor in one campaign

Scale-up
- Central & Southern: same process as PPA, PPB & TRA
- North & Northeast: same process as PPA, PPB & TRA
- Ensure nation-wide adoption

Budgeting
- Budgeting for 2026 driven by risk-based for Central & South
- Budgeting for 2027 driven by risk-based for Central & South
- Ensure nation-wide adoption